# Web Material

# Maternal Medication Use and Childhood Cancer in Offspring – Systematic Review and Considerations for Researchers

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Web Table 1. Example of Search Terms and Search Strategy for the PubMed Database

| **Patient:** Children | **Exposure:** Prenatal exposure | **Outcome:** Cancer |
|---|---|---|
| **MeSH terms** | | |
| Child<br>Child, preschool<br>Adolescent<br>Infant | Prenatal Exposure Delayed Effects<br>Maternal Exposure/*adverse effects<br>Pregnancy Complications/*drug therapy<br>Fetus/*drug effects<br>Maternal-Fetal Exchange | Neoplasms |
| **All fields** | | |
| Child<br>Childhood | Prenatal exposure*<br>Maternal exposure*<br>Pregnancy exposure*<br>Fetal exposure*<br>Foetal exposure*<br>In utero exposure* | Neoplasm*<br>Cancer*<br>Tumor*<br>Tumour*<br>Malignanc*<br>Leukemia*<br>Leukaemia*<br>Lymphoma*<br>Astrocytoma*<br>Neuroblastoma*<br>Nephroblastoma*<br>Rhabdomyosarcoma* |

Search string:

(((((((("prenatal exposure") OR "maternal exposure") OR "pregnancy exposure") OR "fetal exposure") OR "foetal exposure") OR "in utero exposure")) OR (((((((("Prenatal Exposure Delayed Effects"[Mesh]) OR "Maternal Exposure/adverse effects"[Mesh]) OR "Pregnancy Complications/drug therapy"[Mesh])) OR "Fetus/drug effects"[Mesh]) OR "Maternal-Fetal Exchange"[Mesh]))) AND (((((("Child"[Mesh] OR "Child, Preschool"[Mesh]) OR "Infant"[Mesh]) OR "Adolescent"[Mesh])) OR ((child) OR childhood))) AND ((((((((((((neoplasm*) OR cancer*) OR tumor*) OR tumour*) OR malignanc*) OR leukemia*) OR leukaemia*) OR lymphoma*) OR astrocytoma*) OR neuroblastoma*) OR nephroblastoma*) OR rhabdomyosarcoma*)) OR "Neoplasms"[Mesh]))

Web Table 2. Characteristics of the Included Studies by Medication Group

| Reference | Exposure | Outcome | Materials and methods | |
|---|---|---|---|---|
| First author, year | Medication/*exposed*<br>*Ascertainment*<br>Exposure window: during pregnancy, unless otherwise stated | Classification system, type of cancer/*cases* | Design/*sample size*<br>*Study characteristics*<br>Statistical analysis | Method to account for confounding in the main analysis |
| ANALGESICS | | | | |
| Bonaventure, 2015 (1) | Other analgesics and antipyretics (ATC group N02B)/*343*<br>*Routinely collected health data* | Classification: ICD<br>ALL/*725*, AML/*128*, NHL/*83*, HL/*31*, astrocytoma/*100*, medulloblastoma/*48*, rhabdomyosarcoma/*54*, Ewings sarcoma/*20*, neuroblastoma/*78*, nephroblastoma/*60* | Case-control/*4122*<br>*UK, UKCCS, mean age 5.8 (max age 15 years)*<br>Unconditional logistic regression | Matching on sex, month and year of birth, and region of residence. Adjustment for year of birth |
| Bunin, 1994 (2) | Prescription analgesics/*21*<br>*Maternal retrospective report* | Classification system not stated<br>Astrocytoma/*155*<br>PNET/*166* | Case-control/*332*<br>*US and Canada, Children's Cancer Group, ages 0-5 years*<br>Conditional logistic regression | Matching on telephone area code, year of birth and ethnicity. Adjustment for income (only for astrocytoma) |
| Carozza, 1995 (3) | Lidocaine/*1*, Propoxyphene HCL/*15*, Propranolol/*3*, Tetracaine/*1*<br>*Maternal retrospective report* | Classification: Histological Typing of Tumors of the Central Nervous System. International Histological Classification of Tumors<br>Any brain tumors/*361* | Case-control/*1444*<br>*US, SEER, upper age limit 18 years*<br>Conditional logistic regression, for main exposure (any n-nitrosatable drugs), descriptive statistics for medications stated here | Matching on age, sex and maternal ethnicity |
| Cook, 2004 (4) | Acetaminophen/*327*, Aspirin/*24*, Codeine/*28*, Ibuprofen/*28*<br>Each of the above during pregnancy or breastfeeding<br>*Maternal retrospective report* | Classification system not stated<br>Neuroblastoma/*504* | Case-control/*1008*<br>*US and Canada, Children's Cancer Group, upper age limit 18 years*<br>Conditional logistic regression | Matching on age. Adjustment for maternal age, education, and ethnicity |
| Couto, 2015 (5) | Acetaminophen/*83*, Dipyrone/*226*, Acetaminophen and dipyrone/*41*, Acetyl salicylic acid/*25*<br>*Maternal retrospective report, uses controls with other life-threatening illnesses to reduce recall bias* | Classification system not stated<br>ALL/*176*, AML/*55*, acute leukemia with rearrangements in the MLL gene/*43* | Case-control/*642*<br>*Brazil, ages 0-2 years*<br>Unconditional logistic regression | Matching on region. Adjustment for ethnicity, maternal age, education, pesticide exposure, and |

| Reference<br>First author, year | Exposure<br>Medication/*exposed*<br>*Ascertainment*<br>Exposure window: during pregnancy, unless otherwise stated | Outcome<br>Classification system, type of cancer/*cases* | Materials and methods<br>Design/*sample size*<br>*Study characteristics*<br>Statistical analysis | <br>Method to account for confounding in the main analysis |
|---|---|---|---|---|
| | | | | hormone use during pregnancy, and child birth weight |
| Gilman, 1989 (6) | Analgesics and antipyretics/*444*<br>*Maternal retrospective report and medical records. Children are considered exposed if medication use is present in either source* | Classification system not stated<br>Any cancer/*8059* | Case-control/*16,118*<br>*UK, OSCC, age not specified, only that cases are childhood deaths from cancer*<br>Conditional logistic regression | Matching on date of birth, sex and district. Adjustment for abdominal x-rays in pregnancy, birth order among live born siblings, maternal age, and socioeconomic position |
| Grufferman, 1982 (7) | Aspirin/*67*, Analgesics and cold medications/*38*<br>Each of the above 1 year prior to pregnancy or during pregnancy<br>*Maternal retrospective report* | Classification system not stated<br>Rhabdomyosarcoma/*33* | Case-control/*132*<br>*US ages 0-14 years*<br>"Unmatched methods", not specified further | Matching on age, sex, and ethnicity |
| Hartley, 1988 (8) | Analgesics/*not reported*<br>*Maternal retrospective report, validated in prescription data* | Classification system not stated<br>Bone or soft tissue sarcomas/*73* | Case-control/*219*<br>*UK, IRESCC, ages 0-15 years*<br>Mantel-Haenszel, 95% CI by Cornfield's formula | Matching. Matching factors not reported, but they refer to a paper where methods are described (age and sex) |
| Heck, 2015 (9) | Prescription pain medication/*18*<br>*Maternal retrospective report* | Classification system not stated<br>Sporadic retinoblastoma, unilateral/*187*, Sporadic retinoblastoma, bilateral/*95* | Case-control/*426*<br>*US/Canada, ages 0-15 years*<br>Unconditional logistic regression | Matching on year of birth. Adjustment for maternal ethnicity, education, income, age, and smoking in pregnancy |
| Kelty & Hulse, 2017 (10) | Opioid maintenance treatment (methadone, buprenorphine, naltrexone)/*387*<br>*Routinely collected health data* | Classification: ICD<br>Any cancer/*4* | Cohort/*774*<br>*Australia, ages 0-11 years*<br>Descriptive statistics | No adjustment |
| Kramer, 1987 (11) | Prescription analgesics/*not reported*<br>*Maternal retrospective report, validated in medical records for a subsample. Validity compared between cases and controls* | Classification system not stated<br>Neuroblastoma/*104* | Case-control/*205*<br>*US, median age 1 year*<br>90% CI calculated by Miettinen's technique | Matching on telephone area code, ethnicity, and year of birth |
| Kuijten, 1990 (12) | General anesthesia/*not reported*<br>*Maternal retrospective report* | Classification system not stated<br>Astrocytoma/*163* | Case-control/*326*<br>*US, ages 0-14 years* | Matching on telephone area code, ethnicity, and age |

| Reference | Exposure | Outcome | Materials and methods | |
|---|---|---|---|---|
| First author, year | Medication/*exposed*<br>*Ascertainment*<br>Exposure window: during pregnancy, unless otherwise stated | Classification system, type of cancer/*cases* | Design/*sample size*<br>*Study characteristics*<br>Statistical analysis | Method to account for confounding in the main analysis |
| | | | Conditional logistic regression | |
| McCredie, 1994 (13) | Non-prescription analgesics in the month before or during pregnancy/*14*<br>*Maternal retrospective report* | Classification: ICD<br>Brain or cranial nerves tumors/*82* | Case-control/*246*<br>*Australia, ages 0-14 years*<br>Conditional logistic regression | Matching on sex and age.<br>Adjustment for parental education |
| McKinney, 1987 (14) | Analgesics, Narcotic analgesics/*not reported*<br>*Maternal retrospective report and medical records, findings from the two data sources are reported separately* | Classification system not stated<br>Main outcome:<br>Leukemias or lymphomas/*234*<br>Secondary outcomes:<br>Leukemias/*171*<br>Only reported separately for significant associations | Case-control/*702*<br>*UK, IRESCC, ages 0-15 years*<br>Mantel-Haenszel | Matching on age and sex |
| McKinney, 1999 (15) | Analgesics/*36*, General anesthetics/*6*<br>*Routinely collected health data* | Classification: ICCC<br>Main outcome: Leukemia/*144*<br>Secondary outcome: ALL/*124*, lymphomas/*45*, CNS tumors/*75*, other solid tumors/*126* | Case-control/*415*<br>*Scotland, ages 0-14 years*<br>Conditional logistic regression | Matching on age, area of residence, and sex |
| Ognjanovic, 2011 (16) | Aspirin/*59 before, 21 after knowledge of pregnancy*<br>Non-aspirin NSAIDs/*193 before, 58 after knowledge of pregnancy*<br>Acetaminophen/*377 before, 407 after knowledge of pregnancy*<br>Also by any use or regular use<br>*Maternal retrospective report* | Classification system not stated<br>Infant ALL/*262*, infant AML/*172*, also analyzed by MLL gene translocation status | Case-control/*582*<br>*US, Children's Oncology Group, ages 0-1 years*<br>Unconditional logistic regression | Matching on year of birth and region of residence.<br>Adjustment for maternal age, ethnicity, alcohol use during pregnancy, income, and child year of birth |
| Pombo de Oliveira, 2006 (17) | Dipyrone/*352*, Others (Acetaminophen, aspirin, hyoscine, codeine)/*161*<br>Each of the above 3 months before pregnancy, during pregnancy, or during breastfeeding<br>*Maternal retrospective report* | Classification system not stated<br>Infant acute leukemias/*202* | Case-control/*642*<br>*Brazil, ages 0-21 months*<br>Unconditional logistic regression | Matching on age and region.<br>Adjustment for region, sex, birth weight, maternal age, and income |
| Preston-Martin, 1982 (18) | General anesthesia/*6 discordant pairs*<br>*Maternal retrospective report* | Classification system not stated<br>Tumor of the brain or cranial meninges, excluding tumors in infants/*209* | Case-control/*418*<br>*US, ages 0-25 years*<br>Conditional logistic regression | Matching on sex, ethnicity, birth year, and socioeconomic status |

| Reference | Exposure | Outcome | Materials and methods | |
|---|---|---|---|---|
| First author, year | Medication/*exposed*<br>*Ascertainment*<br>Exposure window: during pregnancy, unless otherwise stated | Classification system, type of cancer/*cases* | Design/*sample size*<br>*Study characteristics*<br>Statistical analysis | Method to account for confounding in the main analysis |
| Ross, 2003 (19) | Acetaminophen with codeine/*19*<br>*Routinely collected health data* | Classification system not stated<br>IL/*243*, ALL/*157*, AML/*77* | Case-control/*636*<br>*US, ages 0-18 months*<br>Conditional logistic regression | Matching on birth year and telephone area. Adjustment for maternal age, education and income |
| Salonen, 1976 (20) | Salicylates/*4*, Pyrazol derivatives/*7*, Aniline derivatives/*4*,<br>Euphoriant analgesics/*16*,<br>Medication for migraine and nausea/*23*<br>*Routinely collected health data* | Classification: ICD<br>Main outcomes: Leukemias/*373*, brain tumors/*245*, other tumors/*354*<br>Secondary outcomes: Subgroups of the "other tumors"-category; kidney tumors/*96*, eye tumors/*37*, bone tumors/*56* (only reported for most commonly used medications) | Case-control/*1944*<br>*Finland, ages 0-14 years*<br>MacNemar's test, descriptive for rarely used medications | Matching on date of birth |
| Schüz, 2007 (21) | Analgesics/*142*<br>*Maternal retrospective report from questionnaire data validated through phone interview* | Classification system not stated<br>ALL/*650*, AML/*105*, NHL/*172*, CNS tumors/*399*, neuroblastoma/*157*, nephroblastoma/*147*, bone tumor/*97*, soft tissue sarcoma/*137* | Case-control/*2652*<br>*Germany, ages 0-14 years*<br>Frequency matched conditional logistic regression | Matching on sex, date of birth, and district. Adjustment for age, sex, year of birth, degree of urbanization, and socioeconomic position |
| Sharpe & Franco, 1996 (22) | Dipyrone/*70*, Analgesics (other than NSAIDs)/*145*, NSAIDs/*24*<br>*Maternal retrospective report* | Classification system not stated<br>Nephroblastoma/*109* | Case-control/*327*<br>*Brazil, age range not reported, mean age 3 years and 5 months*<br>Logistic regression | Matching on age, sex, and interviewer, stratified on socioeconomic position |
| Shaw, 2004 (23) | Anti-inflammatories/*4*, Analgesics/*29*<br>*Maternal retrospective report* | Classification: ICD<br>ALL/*789* | Case-control/*1578*<br>*Canada, age 0-14 years*<br>Conditional logistic regression | Matching on sex and age at diagnosis. Adjustment for maternal age and education |
| Shu, 1995 (24) | Pain relievers/*209*<br>*Maternal retrospective report* | Classification system not stated<br>Malignant germ-cell tumors/*105* | Case-control/*744*<br>*US and Canada, CCG (Children's Cancer Group), ages 0-15 years* | Matching on telephone area code. Adjustment for child age, sex, gestational age, number of siblings, and |

| Reference<br>First author, year | Exposure<br>Medication/*exposed*<br>*Ascertainment*<br>Exposure window: during pregnancy, unless otherwise stated | Outcome<br>Classification system, type of cancer/*cases* | Materials and methods<br>Design/*sample size*<br>*Study characteristics*<br>Statistical analysis | <br>Method to account for confounding in the main analysis |
|---|---|---|---|---|
| | | | Unconditional logistic regression | maternal education and smoking |
| Stålberg, 2010 (25) | Any analgesics/*71*, Acetyl salicylic acid or NSAID/*39*, Opioids/*25*, Acetaminophen/*13*<br>*Routinely collected health data* | Classification: ICD<br>Brain tumor (ICD-7 code 193)/*512* | Case-control/*1037*<br>*Sweden, ages 0-15 years*<br>Logistic regression | Matching on sex and birth year. Adjustment for maternal age, parity, country of birth, and level of hospital (primary, secondary, or tertiary) |
| Wang, 2019 (26) | Antipyretic analgesia intake 6 months before or during pregnancy/*99*<br>*Maternal retrospective report* | Classification system not stated<br>ALL/*345* | Case-control/*690*<br>*China, ages 0-15 years*<br>Unconditional logistic regression | Matching on age, sex, and residence region. Adjustment for age, sex, residence region, birth weight, delivery mode, family history of cancer, income, exposure to pesticides (unclear whether this refers to mother or child), parental age, education, paternal exposure to pesticides, paternal cigarette smoking, and alcohol use |
| Wen, 2002 (27) | Anti-inflammatory drugs/*18*<br>Pain relievers/*61*<br>Each of the above in the year before pregnancy, through pregnancy and/or during breastfeeding<br>*Maternal retrospective report* | Classification system not stated<br>ALL/*1842* | Case-control/*3828*<br>*US, ages 0-14 years*<br>Conditional logistic regression | Matching on age, ethnicity, and telephone area. Adjustment for parental income, education, ethnicity, smoking and alcohol use |
| Zanrosso, 2010 (28) | Dipyrone/*102*<br>*Maternal retrospective report* | Classification system not stated<br>IL/*132* | Case-control/*263*<br>*Brazil, ages 0-21 years*<br>Unconditional logistic regression | Matching on age, no adjustment for the association between medication use and IL |
| ANTACIDS | | | | |
| Bonaventure, 2015 (1) | Alimentary tract and metabolism (ATC group A)/*789* | Classification: ICD | Case-control/*4122* | Matching on sex, month and year of birth, and region of |

| Reference<br>First author, year | Exposure<br>Medication/*exposed*<br>*Ascertainment*<br>Exposure window: during pregnancy, unless otherwise stated | Outcome<br>Classification system, type of cancer/*cases* | Materials and methods<br>Design/*sample size*<br>*Study characteristics*<br>Statistical analysis | <br>Method to account for confounding in the main analysis |
|---|---|---|---|---|
| | *Routinely collected health data* | ALL/*725*, AML/*128*, NHL/*83*, HL/*31*, astrocytoma/*100*, medulloblastoma/*48*, rhabdomyosarcoma/*54*, Ewings sarcoma/*20*, neuroblastoma/*78*, nephroblasoma/*60* | *UK, UKCCS, mean age 5.8 (max age 15 years)*<br>Unconditional logistic regression | residence. Adjustment for year of birth |
| Gilman, 1989 (6) | Antacids/*150*<br>*Maternal retrospective report and medical records. Children are considered exposed if medication use is present in either source* | Classification system not stated<br>Any cancer/*8059* | Case-control/*16,118*<br>*UK, OSCC, age not specified, only that cases are childhood deaths from cancer*<br>Conditional logistic regression | Matching on date of birth, sex and district. Adjustment for abdominal x-rays in pregnancy, birth order among live born siblings, maternal age, and socioeconomic position |
| Hartley, 1988 (8) | Antacids/*not reported*<br>*Maternal retrospective report, validated in prescription data* | Classification system not stated<br>Bone or soft tissue sarcomas/*73* | Case-control/*219*<br>*UK, IRESCC, ages 0-15 years*<br>Mantel-Haenszel, 95% CI by Cornfield's formula | Matching. Matching factors not reported, but they refer to a paper where methods are described (age and sex) |
| McKinney, 1999 (15) | Antacids/*28*<br>*Routinely collected health data* | Classification: ICCC<br>Main outcome: Leukemia/*144*<br>Secondary outcome: ALL/*124*, lymphomas/*45*, CNS tumors/*75*, other solid tumors/*126* | Case-control/*415*<br>*Scotland, ages 0-14 years*<br>Conditional logistic regression | Matching on age, area of residence, and sex |
| Michalek, 1996 (29) | Antacids/*116*<br>*Maternal retrospective report* | Classification system not stated<br>Neuroblastoma/*183* | Case-control/*555*<br>*US, ages 0-14 years*<br>Unconditional logistic regression | Matching on year of birth |
| Salonen, 1976 (20) | Antacids/*27*<br>*Routinely collected health data* | Classification: ICD<br>Main outcomes: Leukemias/*373*, brain tumors/*245*, other tumors/*354*<br>Secondary outcomes: Subgroups of the "other tumors"-category; kidney tumors/*96*, eye tumors/*37*, | Case-control/*1944*<br>*Finland, ages 0-14 years*<br>MacNemar's test, descriptive for rarely used medications | Matching on date of birth |

| Reference<br>First author, year | Exposure<br>Medication/*exposed*<br>*Ascertainment*<br>Exposure window: during pregnancy, unless otherwise stated | Outcome<br>Classification system, type of cancer/*cases* | Materials and methods<br>Design/*sample size*<br>*Study characteristics*<br>Statistical analysis | <br>Method to account for confounding in the main analysis |
|---|---|---|---|---|
| | | bone tumors/*56* (only reported for most commonly used medications) | | |
| Shaw, 2004 (23) | Antacids/*61*<br>*Maternal retrospective report* | Classification: ICD<br>ALL/*789* | Case-control/*1578*<br>*Canada, age 0-14 years*<br>Conditional logistic regression | Matching on sex and age at diagnosis. Adjustment for maternal age and education |
| Stålberg, 2010 (25) | Alimentary tract (ATC group A), including mainly antacid and laxatives/*102*<br>*Routinely collected health data* | Classification: ICD<br>Brain tumor (ICD-7 code 193)/*512* | Case-control/*1037*<br>*Sweden, ages 0-15 years*<br>Logistic regression | Matching on sex and birth year. Adjustment for maternal age, parity, country of birth, and level of hospital (primary, secondary, or tertiary) |
| ANTIDIABETICS | | | | |
| Gilman, 1989 (6) | Insulin/*13*<br>*Maternal retrospective report and medical records. Children are considered exposed if medication use is present in either source* | Classification system not stated<br>Any cancer/*8059* | Case-control/*16,118*<br>*UK, OSCC, age not specified, only that cases are childhood deaths from cancer*<br>Conditional logistic regression | Matching on date of birth, sex and district. Adjustment for abdominal x-rays in pregnancy, birth order among live born siblings, maternal age, and socioeconomic position |
| Ognjanovic, 2009 (30) | Insulin/*8*<br>*Maternal retrospective report* | Classification system not stated<br>Main outcome: Leukemias/*158*<br>Secondary outcomes: ALL/*97*, AML/*61* | Case-control/*331*<br>*US, Children's Oncology Group (US), ages 0-19 years, all cases and controls have Down syndrome*<br>Descriptive statistics | Matching on age and sex |
| Seppälä, 2020 (31) | Insulin only/*127*<br>Metformin only/*48*<br>Insulin and metformin/*12*<br>*Routinely collected health data* | Classification: ICCC<br>Main outcome: Any cancer/*2029*<br>Secondary outcomes: Leukemias/*649*, lymphomas/*149*, CNS tumors/*484*, other cancers/*743* (not reported for medication exposure) | Case-cohort/*12,132*<br>*Finland, ages 0-19 years*<br>Conditional logistic regression | Adjustment for maternal age, parity, and smoking |

| Reference<br>First author, year | Exposure<br>Medication/*exposed*<br>*Ascertainment*<br>Exposure window: during pregnancy, unless otherwise stated | Outcome<br>Classification system, type of cancer/*cases* | Materials and methods<br>Design/*sample size*<br>*Study characteristics*<br>Statistical analysis | Method to account for confounding in the main analysis |
|---|---|---|---|---|
| Shaw, 2004 (23) | Diabetes medications/*9*<br>*Maternal retrospective report* | Classification: ICD<br>ALL/*789* | Case-control/*1578*<br>*Canada, age 0-14 years*<br>Conditional logistic regression | Matching on sex and age at diagnosis. Adjustment for maternal age and education |
| Søegaard, 2018 (32) | Insulin/*3857*<br>*Routinely collected health data* | Classification: ICD<br>ALL/*492* | Cohort study/*1,187,482*<br>*Denmark, age 0-15 years*<br>Cox proportional hazards models | No adjustment for the analysis on medication exposure |
| ANTIHISTAMINES | | | | |
| Buckley, 1989 (33) | Antihistamines in the year before pregnancy or during pregnancy/*9*<br>*Maternal retrospective report* | Classification system not stated<br>Hepatoblastoma/*75* | Case-control/*150*<br>*US and Canada, CCSG, upper age limit not stated, but the majority of children were under 2 years of age*<br>Conditional logistic regression | Matching on age |
| Bunin, 1989 (34) | Antihistamines/*30*<br>*Maternal retrospective report* | Classification system not stated<br>Nonheritable retinoblastoma/*115*<br>Sporadic heritable retinoblastoma (bilateral cases or unilateral cases with 13q chromosomal deletion)/*67*<br>Cases with family history were excluded | Case-control/*383*<br>*US and Canada, CCSG, upper age limit not stated*<br>Conditional logistic regression | Matching on telephone area code, year of birth and ethnicity. Adjustment for paternal education |
| Bunin, 1994 (2) | Antihistamines/*41*<br>*Maternal retrospective report* | Classification system not stated<br>Astrocytoma/*155*<br>PNET/*166* | Case-control/*332*<br>*US and Canada, Children's Cancer Group, ages 0-5 years*<br>Conditional logistic regression | Matching on telephone area code, year of birth and ethnicity. Adjustment for income (only for astrocytoma) |
| Carozza, 1995 (3) | Chlorpheniramine/*2*, Chlorpheniramine maleate/*17*, Cimetidine/*3*, Promethazine HCL/*1*<br>*Maternal retrospective report* | Classification: Histological Typing of Tumors of the Central Nervous System. International Histological Classification of Tumors | Case-control/*1444*<br>*US, SEER, upper age limit 18 years*<br>Conditional logistic regression, for main exposure (any n- | Matching on age, sex and maternal ethnicity |

| Reference | Exposure | Outcome | Materials and methods | |
|---|---|---|---|---|
| First author, year | Medication/*exposed*<br>*Ascertainment*<br>Exposure window: during pregnancy, unless otherwise stated | Classification system, type of cancer/*cases* | Design/*sample size*<br>*Study characteristics*<br>Statistical analysis | Method to account for confounding in the main analysis |
| | | Any brain tumors/*361* | nitrosatable drugs), descriptive statistics for medications stated here | |
| Cardy, 2006 (35) | Antihistamines/*204*<br>*Maternal retrospective report* | Classification: ICD-O<br>Main outcome:<br>Any brain tumor/*1218*<br>Secondary outcomes:<br>Astroglial/*623*, PNET/*259*, other glial/*327* by morphological codes | Case-control/*3441*<br>*Europe, North America, Australia, SEARCH, upper age limit 19 years*<br>Unconditional logistic regression | Matching, not further specified. Adjustment for center, age, sex, year of birth, and parental education. Vitamin supplementation (unclear whether in mothers or children) was considered as both a confounder and effect modifier |
| Cook, 2004 (4) | Chlorpheniramine/*44*, Diphenhydramine/*18*, Doxylamine/*11*, Brompheniramine/*12*, Promethazine/*11*<br>Each during pregnancy or breastfeeding<br>*Maternal retrospective report* | Classification system not stated<br>Neuroblastoma/*504* | Case-control/*1008*<br>*US and Canada, Children's Cancer Group, upper age limit 18 years*<br>Conditional logistic regression | Matching on age. Adjustment for maternal age, education, and ethnicity |
| Cordier, 1994 (36) | Antihistamines/*7*<br>*Maternal retrospective report* | Classification system not stated<br>Brain tumors/*75* | Case-control/*188*<br>*France, ages 0-15 years*<br>Conditional logistic regression | Matching on birth year. Adjustment for age, sex and maternal age |
| Kuijten, 1990 (12) | Antihistamines/*not reported*<br>*Maternal retrospective report* | Classification system not stated<br>Astrocytoma/*163* | Case-control/*326*<br>*US, ages 0-14 years*<br>Conditional logistic regression | Matching on telephone area code, ethnicity, and age |
| McCredie, 1994 (13) | Oral antihistamines in the month before or during pregnancy/*7*<br>*Maternal retrospective report* | Classification: ICD<br>Brain or cranial nerves tumors/*82* | Case-control/*246*<br>*Australia, ages 0-14 years*<br>Conditional logistic regression | Matching on sex and age. Adjustment for parental education |
| McKinney, 1987 (14) | Antihistamines/*not reported*<br>*Maternal retrospective report and medical records, findings from the two data sources are reported separately* | Classification system not stated<br>Main outcome:<br>Leukemias/lymphomas/*234*<br>Secondary outcomes:<br>Leukemias/*171*<br>Only reported separately for significant associations | Case-control/*702*<br>*UK, IRESCC, ages 0-15 years*<br>Mantel-Haenszel | Matching on age and sex |

| Reference | Exposure | Outcome | Materials and methods | |
|---|---|---|---|---|
| First author, year | Medication/*exposed*<br>*Ascertainment*<br>Exposure window: during pregnancy, unless otherwise stated | Classification system, type of cancer/*cases* | Design/*sample size*<br>*Study characteristics*<br>Statistical analysis | Method to account for confounding in the main analysis |
| Preston-Martin, 1982 (18) | Antihistamines/*31 discordant pairs*<br>*Maternal retrospective report* | Classification system not stated<br>Tumor of the brain or cranial meninges, excluding tumors in infants/*209* | Case-control/*418*<br>*US, ages 0-25 years*<br>Conditional logistic regression | Matching on sex, ethnicity, birth year, and socioeconomic status |
| Ross, 2003 (19) | Diphenhydramine/*11* , Promethazine/*23*<br>*Routinely collected health data* | Classification system not stated<br>IL/*243*, ALL/*157*, AML/*77* | Case-control/*636*<br>*US, ages 0-18 months*<br>Conditional logistic regression | Matching on birth year and telephone area. Adjustment for maternal age, education and income |
| Robison, 1989 (37) | Antihistamine in the year before pregnancy, through pregnancy and/or during breastfeeding/*38*<br>Also analyses by duration and frequency<br>*Maternal retrospective report, validated by comparisons to associations in four other independent case-control studies where cases of different cancers did not report more medication use than controls* | Classification system not stated<br>ANLL/*204* | Case-control/*408*<br>*US, ages 0-18 years*<br>Conditional logistic regression | Matching on date of birth, ethnicity, and telephone area code, stated that they adjust for confounders, but these are not specified |
| Sharpe & Franco, 1996 (22) | Antiallergics/*3*<br>*Maternal retrospective report* | Classification system not stated<br>Nephroblastoma/*109* | Case-control/*327*<br>*Brazil, age range not reported, mean age 3 years and 5 months*<br>Logistic regression | Matching on age, sex, and interviewer, stratified on socioeconomic position |
| Shaw, 2004 (23) | Antihistamine/*11*<br>*Maternal retrospective report* | Classification: ICD<br>ALL/*789* | Case-control/*1578*<br>*Canada, age 0-14 years*<br>Conditional logistic regression | Matching on sex and age at diagnosis. Adjustment for maternal age and education |
| Stålberg, 2010 (25) | Antihistamines/*100*<br>*Routinely collected health data* | Classification: ICD<br>Brain tumor (ICD-7 code 193)/*512* | Case-control/*1037*<br>*Sweden, ages 0-15 years*<br>Logistic regression | Matching on sex and birth year. Adjustment for maternal age, parity, country of birth, and level of hospital (primary, secondary, or tertiary) |
| Wen, 2002 (27) | Antihistamines or allergy remedies in the year before pregnancy, through pregnancy and/or during breastfeeding/*125*<br>*Maternal retrospective report* | Classification system not stated<br>ALL/*1842* | Case-control/*3828*<br>*US, ages 0-14 years*<br>Conditional logistic regression | Matching on age, ethnicity, and telephone area. Adjustment for parental |

| Reference | Exposure | Outcome | Materials and methods | |
|---|---|---|---|---|
| First author, year | Medication/*exposed*<br>*Ascertainment*<br>Exposure window: during pregnancy, unless otherwise stated | Classification system, type of cancer/*cases* | Design/*sample size*<br>*Study characteristics*<br>Statistical analysis | Method to account for confounding in the main analysis |
| | | | | income, education, ethnicity, smoking and alcohol use |
| ANTIHYPERTENSIVES | | | | |
| Buckley, 1989 (33) | Antihypertensives/*1*, Diuretics/*10*<br>Each in the year before pregnancy or during pregnancy<br>*Maternal retrospective report* | Classification system not stated<br>Hepatoblastoma/*75* | Case-control/*150*<br>*US and Canada, CCSG, upper age limit not stated, but the majority of children were under 2 years of age*<br>Conditional logistic regression | Matching on age |
| Bunin, 1987 (38) | Diuretics in the year prior to birth/*10*<br>*Maternal retrospective report* | Classification system not stated<br>Nephroblastoma/*88* | Case-control/*176*<br>*US, ages 0-15 years*<br>Conditional logistic regression | Matching on telephone area code, year of birth and ethnicity. Adjustment for demographic characteristics (not specified further) |
| Cardy, 2006 (35) | Diuretics/*89*<br>*Maternal retrospective report* | Classification: ICD-O<br>Main outcome:<br>Any brain tumor/*1218*<br>Secondary outcomes:<br>Astroglial/*623*, PNET/*259*, other glial/*327* by morphological codes | Case-control/*3441*<br>*Europe, North America, Australia, SEARCH, upper age limit 19 years*<br>Unconditional logistic regression | Matching, not further specified. Adjustment for center, age, sex, year of birth, and parental education. Vitamin supplementation (unclear whether in mothers or children) was considered as both a confounder and effect modifier |
| Carozza, 1995 (3) | Furosemide/*1*, Hydrochlorothiazide/*2*<br>*Maternal retrospective report* | Classification: Histological Typing of Tumors of the Central Nervous System. International Histological Classification of Tumors<br>Any brain tumors/*361* | Case-control/*1444*<br>*US, SEER, upper age limit 18 years*<br>Conditional logistic regression, for main exposure (any n-nitrosatable drugs), descriptive statistics for medications stated here | Matching on age, sex and maternal ethnicity |

| Reference | Exposure | Outcome | Materials and methods | |
|---|---|---|---|---|
| First author, year | Medication/*exposed*<br>*Ascertainment*<br>Exposure window: during pregnancy, unless otherwise stated | Classification system, type of cancer/*cases* | Design/*sample size*<br>*Study characteristics*<br>Statistical analysis | Method to account for confounding in the main analysis |
| Gilman, 1989 (6) | Antihypertensives/*88*, Diuretics/*81*<br>*Maternal retrospective report and medical record. Children are considered exposed if medication use is present in either source* | Classification system not stated<br>Any cancer/*8059* | Case-control/*16,118*<br>*UK, OSCC, age not specified, only that cases are childhood deaths from cancer*<br>Conditional logistic regression | Matching on date of birth, sex and district. Adjustment for abdominal x-rays in pregnancy, birth order among live born siblings, maternal age, and socioeconomic position |
| Hartley, 1988 (8) | Diuretics/*not reported*<br>*Maternal retrospective report, validated in prescription data* | Classification system not stated<br>Bone or soft tissue sarcomas/*73* | Case-control/*219*<br>*UK, IRESCC, ages 0-15 years*<br>Mantel-Haenszel, 95% CI by Cornfield's formula | Matching. Matching factors not reported, but they refer to a paper where methods are described (age and sex) |
| Kramer, 1987 (11) | Diuretics/*not reported*<br>*Maternal retrospective report, validated in medical records for a subsample. Validity compared between cases and controls* | Classification system not stated<br>Neuroblastoma/*104* | Case-control/*205*<br>*US, median age 1 year*<br>90% CI calculated by Miettinen's technique | Matching on telephone area code, ethnicity, and year of birth |
| Kuijten, 1990 (12) | Diuretics/*not reported*<br>*Maternal retrospective report* | Classification system not stated<br>Astrocytoma/*163* | Case-control/*326*<br>*US, ages 0-14 years*<br>Conditional logistic regression | Matching on telephone area code, ethnicity, and age |
| McCredie, 1994 (13) | Diuretics in the month before or during pregnancy/*5*<br>*Maternal retrospective report* | Classification: ICD<br>Brain or cranial nerves tumors/*82* | Case-control/*246*<br>*Australia, ages 0-14 years*<br>Conditional logistic regression | Matching on sex and age. Adjustment for parental education |
| McKinney, 1987 (14) | Diuretics/*not reported*<br>*Maternal retrospective report and medical records, findings from the two data sources are reported separately* | Classification system not stated<br>Main outcome:<br>Leukemias/lymphomas/*234*<br>Secondary outcomes:<br>Leukemias/*171*<br>Only reported separately for significant associations | Case-control/*702*<br>*UK, IRESCC, ages 0-15 years*<br>Mantel-Haenszel | Matching on age and sex |
| Michalek, 1996 (29) | Diuretics/*11*<br>*Maternal retrospective report* | Classification system not stated<br>Neuroblastoma/*183* | Case-control/*555*<br>*US, ages 0-14 years*<br>Unconditional logistic regression | Matching on year of birth |

| Reference | Exposure | Outcome | Materials and methods | |
|---|---|---|---|---|
| First author, year | Medication/*exposed*<br>*Ascertainment*<br>Exposure window: during pregnancy, unless otherwise stated | Classification system, type of cancer/*cases* | Design/*sample size*<br>*Study characteristics*<br>Statistical analysis | Method to account for confounding in the main analysis |
| Olshan, 1993 (39) | Diuretics/*13*<br>Blood pressure medication/*4*<br>*Maternal retrospective report* | Classification system not stated<br>Nephroblastoma/*200* | Case-control/*433*<br>*US, ages 0-15 years*<br>Conditional logistic regression | Matching on age and geographic area. Adjustment for income, and paternal education |
| Preston-Martin, 1982 (18) | Diuretics/*39 discordant pairs*<br>*Maternal retrospective report* | Classification system not stated<br>Tumor of the brain or cranial meninges, excluding tumors in infants/*209* | Case-control/*418*<br>*US, ages 0-25 years*<br>Conditional logistic regression | Matching on sex, ethnicity, birth year, and socioeconomic status |
| Robison, 1989 (37) | Diuretics/*39*, Blood pressure/*4*<br>Each of the above in the year before pregnancy, through pregnancy and/or during breastfeeding. Also analyses by duration and frequency<br>*Maternal retrospective report, validated by comparisons to associations in four other independent case-control studies where cases of different cancers did not report more medication use than controls* | Classification system not stated<br>ANLL/*204* | Case-control/*408*<br>*US, ages 0-18 years*<br>Conditional logistic regression | Matching on date of birth, ethnicity, and telephone area code, stated that they adjust for confounders, but these are not specified |
| Salonen, 1976 (20) | Diuretics/*122*, Antihypertensives/*25*<br>*Routinely collected health data* | Classification: ICD<br>Main outcomes: Leukemias/*373*, brain tumors/*245*, other tumors/*354*<br>Secondary outcomes: Subgroups of the "other tumors"-category; kidney tumors/*96*, eye tumors/*37*, bone tumors/*56* (only reported for most commonly used medications) | Case-control/*1944*<br>*Finland, ages 0-14 years*<br>MacNemar's test, descriptive for rarely used medications | Matching on date of birth |
| Schüz, 2007 (21) | Antihypertensives/diuretics/*40*<br>*Maternal retrospective report from questionnaire data validated through phone interview* | Classification system not stated<br>ALL/*650*, AML/*105*, NHL/*172*, CNS tumors/*399*, neuroblastoma/*157*, nephroblastoma/*147*, bone tumor/*97*, soft tissue sarcoma/*137* | Case-control/*2652*<br>*Germany, ages 0-14 years*<br>Frequency matched conditional logistic regression | Matching on sex, date of birth, and district. Adjustment for age, sex, year of birth, degree of urbanization, and socioeconomic position |

| Reference<br>First author, year | Exposure<br>Medication/*exposed*<br>*Ascertainment*<br>Exposure window: during pregnancy, unless otherwise stated | Outcome<br>Classification system, type of cancer/*cases* | Materials and methods<br>Design/*sample size*<br>*Study characteristics*<br>Statistical analysis | Method to account for confounding in the main analysis |
|---|---|---|---|---|
| Sharpe & Franco, 1996 (22) | Antihypertensives/*19*<br>*Maternal retrospective report* | Classification system not stated<br>Nephroblastoma/*109* | Case-control/*327*<br>*Brazil, age range not reported, mean age 3 years and 5 months*<br>Logistic regression | Matching on age, sex, and interviewer, stratified on socioeconomic position |
| Shu, 1995 (24) | Blood pressure/*6*<br>*Maternal retrospective report* | Classification system not stated<br>Malignant germ-cell tumors/*105* | Case-control/*744*<br>*US and Canada, CCG (Children's Cancer Group), ages 0-15 years*<br>Unconditional logistic regression | Matching on telephone area code. Adjustment for child age, sex, gestational age, number of siblings, maternal education, and smoking |
| Stålberg, 2010 (25) | Antihypertensive/*25*, α-blockers/*16*, β-blockers/*12*, Diuretics/*85*<br>*Routinely collected health data* | Classification: ICD<br>Brain tumor (ICD-7 code 193)/*512* | Case-control/*1037*<br>*Sweden, ages 0-15 years*<br>Logistic regression | Matching on sex and birth year. Adjustment for maternal age, parity, country of birth, and level of hospital (primary, secondary, or tertiary) |
| Wen, 2002 (27) | Diuretics or Lasix/*27*, Blood pressure tablets/*26*<br>Each of the above in the year before pregnancy, through pregnancy and/or during breastfeeding<br>*Maternal retrospective report* | Classification system not stated<br>ALL/*1842* | Case-control/*3828*<br>*US, ages 0-14 years*<br>Conditional logistic regression | Matching on age, ethnicity, and telephone area. Adjustment for parental income, education, ethnicity, smoking and alcohol use |
| ANTI-INFECTIVES | | | | |
| Bonaventure, 2015 (1) | Systemic antibacterials (ATC group J01)/*942*, Penicillins (ATC group J01C)/*846*<br>*Routinely collected health data* | Classification: ICD<br>ALL/*725*, AML/*128*, NHL/*83*, HL/*31*, astrocytoma/*100*, medulloblastoma/*48*, rhabdomyosarcoma/*54*, Ewings sarcoma/*20*, neuroblastoma/*78*, nephroblasoma/*60* | Case-control/*4122*<br>*UK, UKCCS, mean age 5.8 (max age 15 years)*<br>Unconditional logistic regression | Matching on sex, month and year of birth, and region of residence. Adjustment for year of birth |
| Buckley, 1989 (33) | Antibiotics in the year before pregnancy or during pregnancy/*31*<br>*Maternal retrospective report* | Classification system not stated<br>Hepatoblastoma/*75* | Case-control/*150*<br>*US and Canada, CCSG, upper age limit not stated, but the* | Matching on age |

| Reference | Exposure | Outcome | Materials and methods | |
|---|---|---|---|---|
| First author, year | Medication/*exposed*<br>*Ascertainment*<br>Exposure window: during pregnancy, unless otherwise stated | Classification system, type of cancer/*cases* | Design/*sample size*<br>*Study characteristics*<br>Statistical analysis | Method to account for confounding in the main analysis |
| | | | *majority of children were under 2 years of age*<br>Conditional logistic regression | |
| Bunin, 1994 (2) | Medication against vaginal infection/*59*<br>*Maternal retrospective report* | Classification system not stated<br>Astrocytoma/*155*<br>PNET/*166* | Case-control/*332*<br>*US and Canada, Children's Cancer Group, ages 0-5 years*<br>Conditional logistic regression | Matching on telephone area code, year of birth and ethnicity. Adjustment for income (only for astrocytoma) |
| Carozza, 1995 (3) | Ampicillin/*13*, Ampicillin trihydrate/*1*, Doxycline/*2*, Erythromycin/*4*, Oxytetracycline/*2*, Penicillin V potassium/*7*, Tetracycline HCL/*1*<br>*Maternal retrospective report* | Classification: Histological Typing of Tumors of the Central Nervous System. International Histological Classification of Tumors<br>Any brain tumors/*361* | Case-control/*1444*<br>*US, SEER, upper age limit 18 years*<br>Conditional logistic regression, for main exposure (any n-nitrosatable drugs), descriptive statistics for medications stated here | Matching on age, sex and maternal ethnicity |
| Cook, 2004 (4) | Amoxicillin/*102*, Ampicillin/*21*, Cephalexin/*17*, Erythromycin/*27*, Nitrofurantoin/*5*, Penicillin/*23*, Sulfamethoxazole/*14*, Trimethoprim/*15*<br>Each during pregnancy or breastfeeding<br>*Maternal retrospective report* | Classification system not stated<br>Neuroblastoma/*504* | Case-control/*1008*<br>*US and Canada, Children's Cancer Group, upper age limit 18 years*<br>Conditional logistic regression | Matching on age. Adjustment for maternal age, education, and ethnicity |
| Gilman, 1989 (6) | Antibiotics/*631*, Antimalarials/*16*<br>Antituberculosis drugs/*13*, Sulphonamides/*116*, Vaccines/*235*<br>*Maternal retrospective report and medical records. Children are considered exposed if medication use is present in either source* | Classification system not stated<br>Any cancer/*8059* | Case-control/*16,118*<br>*UK, OSCC, age not specified, only that cases are childhood deaths from cancer*<br>Conditional logistic regression | Matching on date of birth, sex and district. Adjustment for abdominal x-rays in pregnancy, birth order among live born siblings, maternal age, and socioeconomic position |
| Gradel & Kaerlev, 2015 (40) | Antibiotics/*1201*<br>Secondary exposures:<br>Narrow spectrum β-lactams/*714*,<br>Broad spectrum β-lactams/*715*, | Classification: ICD<br>Main outcome:<br>Any leukemias/*360*<br>Secondary outcome: | Case-control/*3869*<br>*Denmark, ages 0-15 years*<br>Conditional logistic regression | Matching on age, sex, and municipality. Adjustment for birth weight, birth order, maternal age, migration, |

| Reference<br>First author, year | Exposure<br>Medication/*exposed*<br>*Ascertainment*<br>Exposure window: during pregnancy, unless otherwise stated | Outcome<br>Classification system, type of cancer/*cases* | Materials and methods<br>Design/*sample size*<br>*Study characteristics*<br>Statistical analysis | <br>Method to account for confounding in the main analysis |
|---|---|---|---|---|
| | Sulphonamides or trimethoprim*/317*, Macrolides*/147*<br>Also analysis by number of treatments<br>*Routinely collected health data* | ALL/*284* | | education, labor market affiliation, and income |
| Grufferman, 1982 (7) | Antibiotics 1 year prior to or during pregnancy/*32*<br>*Maternal retrospective report* | Classification system not stated<br>Rhabdomyosarcoma/*33* | Case-control/*132*<br>*US, ages 0-14 years*<br>"Unmatched methods", not specified further | Matching on age, sex, and ethnicity |
| Hartley, 1988 (8) | Urinary anti-infectives/*not reported*<br>*Maternal retrospective report, validated in prescription data* | Classification system not stated<br>Bone or soft tissue sarcomas/*73* | Case-control/*219*<br>*UK, IRESCC, ages 0-15 years*<br>Mantel-Haenszel, 95% CI by Cornfield's formula | Matching. Matching factors not reported, but they refer to a paper where methods are described (age and sex) |
| Heck, 2015 (9) | Antibiotics/*53*<br>*Maternal retrospective report* | Classification system not stated<br>Sporadic retinoblastoma, unilateral/*187*, Sporadic retinoblastoma, bilateral/*95* | Case-control/*426*<br>*US/Canada, ages 0-15 years*<br>Unconditional logistic regression | Matching on year of birth. Adjustment for maternal ethnicity, education, income, age, and smoking in pregnancy |
| Infante-Rivard, 2000 (41) | Antibiotics/*118*<br>*Maternal retrospective report* | Classification: ICD<br>ALL/*491* | Case-control/*982*<br>*Canada, upper age limit 10 years*<br>Conditional logistic regression | Matching on sex, age, and area of residence. Adjustment for maternal age and education |
| Kaatsch, 2010 (42) | Antibiotics/*224*<br>*Maternal retrospective report* | Classification: ICCC<br>Main outcome: Any cancer/*1867*<br>Secondary outcomes: ALL/*650*, AML/*105*, Burkitt lymphoma/*12*, NHL/*141*, medulloblastoma/*94*, astrocytoma/*92*, ependymoma/*44*, neuroblastoma/*160*, nephroblastoma/*147*, bone tumor/*97*, soft tissue sarcoma/*137* | Case-control/*3924*<br>*Germany, 0-14 years*<br>Conditional logistic regression | Matching on community, sex, and year of birth. Adjustment for sex, age, degree of urbanization, and socioeconomic position |
| Kumar, 2014 (43) | Chloramphenicol/*not reported*<br>*Maternal retrospective report* | Classification system not stated<br>ALL or AML/*132* | Case-control/*264*<br>*India, ages 0-17 years*<br>z-test for two proportions | Matching on age, sex, and residency |

| Reference | Exposure | Outcome | Materials and methods | |
|---|---|---|---|---|
| First author, year | Medication/*exposed*<br>*Ascertainment*<br>Exposure window: during pregnancy, unless otherwise stated | Classification system, type of cancer/*cases* | Design/*sample size*<br>*Study characteristics*<br>Statistical analysis | Method to account for confounding in the main analysis |
| Kwan, 2007 (44) | Antibiotics 3 months before and/or during pregnancy/*128*<br>*Maternal retrospective report* | Classification: ICD-O<br>Main outcome: Leukemia/*365*<br>Secondary outcome: ALL/*311* | Case-control/*825*<br>*US, NCCLS, ages 0-14 years*<br>Conditional logistic regression | Matching on date of birth, sex, ethnicity (of child and mother), and for part of the recruitment also county of residence at birth. Adjustment for maternal age, income, and education |
| McKinney, 1987 (14) | Antibiotics, Antifungals/*not reported*<br>*Maternal retrospective report and medical records, findings from the two data sources are reported separately* | Classification system not stated<br>Main outcome: Leukemias or lymphomas/*234*<br>Secondary outcomes: Leukemias/*171*<br>Only reported separately for significant associations | Case-control/*702*<br>*UK, IRESCC, ages 0-15 years*<br>Mantel-Haenszel | Matching on age and sex |
| McKinney, 1999 (15) | Antibiotics/*60*, Topical antifungals/*29*<br>*Routinely collected health data* | Classification: ICCC<br>Main outcome: Leukemia/*144*<br>Secondary outcome: ALL/*124*, lymphomas/*45*, CNS tumors/*75*, other solid tumors/*126* | Case-control/*415*<br>*Scotland, ages 0-14 years*<br>Conditional logistic regression | Matching on age, area of residence, and sex |
| Michalek, 1996 (29) | Medications for urinary tract infection/*26*, Medications for bladder infection/*12*<br>*Maternal retrospective report* | Classification system not stated<br>Neuroblastoma/*183* | Case-control/*555*<br>*US, ages 0-14 years*<br>Unconditional logistic regression | Matching on year of birth |
| Momen, 2015 (45) | Main exposure: Antibiotics/*506,194*<br>Secondary exposures: Antibiotics on 3rd ATC-level, selected antibiotics on 5$^{th}$ ATC-level/*not reported*<br>Also analysis by dose<br>*Routinely collected health data* | Classification: ICD<br>Main outcome: Any cancer/*1479*<br>Secondary outcomes: Leukemias/*591*, ALL/*444*, AML/*74*, HL/*23*, NHL/*20*, CNS or SNS tumors/*249*, renal tumors/*121*, hepatic tumors/*29*, endocrine tumors/*93*, testis cancer/*21*, eye cancers/*77*, bone cancers/*44* | Cohort/*1,442,114*<br>*Denmark and Sweden, ages 0-14 years*<br>Cox regression | Adjustment for maternal age, parity, education, smoking during pregnancy, and child country of birth |

| Reference | Exposure | Outcome | Materials and methods | |
|---|---|---|---|---|
| First author, year | Medication/*exposed*<br>*Ascertainment*<br>Exposure window: during pregnancy, unless otherwise stated | Classification system, type of cancer/*cases* | Design/*sample size*<br>*Study characteristics*<br>Statistical analysis | Method to account for confounding in the main analysis |
| Naumburg, 2002 (46) | Local treatment for vaginal infections/*88*, Oral antibiotics/*76*<br>*Routinely collected health data* | Classification: ICD<br>Main outcome: Leukemia/*652*<br>Secondary outcomes: lymphoid leukemias/*578*, myeloid leukemias/*74* (not reported for medication exposures) | Case-cohort/*1304*<br>*Sweden, age 0-16 years*<br>Conditional logistic regression | Matching on sex, birth year and month. Adjustment for maternal age, parity, mode of delivery, smoking, time from membrane rupture until delivery, gestational age, birth weight, and plurality |
| Olshan, 1993 (39) | Antibiotics/*58*<br>*Maternal retrospective report* | Classification system not stated<br>Nephroblastoma/*200* | Case-control/*433*<br>*US, ages 0-15 years*<br>Conditional logistic regression | Matching on age and geographic area. Adjustment for income, and paternal education |
| Pombo de Oliveira, 2006 (17) | Amoxicillin/*87*, Ciprofloxacin/*16*<br>Antifungal (metronidazole)/*16*<br>Each of the above 3 months before pregnancy, during pregnancy, or during breastfeeding<br>*Maternal retrospective report* | Classification system not stated<br>Infant acute leukemias/*202* | Case-control/*642*<br>*Brazil, ages 0-21 months*<br>Unconditional logistic regression | Matching on age and region. Adjustment for region, sex, income, maternal age, and birth weight |
| Robison, 1989 (37) | Antibiotics in the year before pregnancy, through pregnancy and/or during breastfeeding/*97*<br>Also analyses by duration and frequency<br>*Maternal retrospective report, validated by comparisons to associations in four other independent case-control studies where cases of different cancers did not report more medication use than controls* | Classification system not stated<br>ANLL/*204* | Case-control/*408*<br>*US, ages 0-18 years*<br>Conditional logistic regression | Matching on date of birth, ethnicity, and telephone area code, stated that they adjust for confounders, but these are not specified |
| Rodvall, 1990 (47) | Drugs for infections (unspecified)/*11*<br>*Routinely collected health data* | Classification system not stated<br>Any cancer/*95* | Case-control/*285*<br>*Sweden, ages 0-16 years*<br>Multiple conditional logistic regression | Matching on sex, birth year and month. Adjustment for maternal age, drug use, obstetric complications, previous miscarriage, socioeconomic position and child gestational age. |

| Reference<br>First author, year | Exposure<br>Medication/*exposed*<br>*Ascertainment*<br>Exposure window: during pregnancy, unless otherwise stated | Outcome<br>Classification system, type of cancer/*cases* | Materials and methods<br>Design/*sample size*<br>*Study characteristics*<br>Statistical analysis | <br>Method to account for confounding in the main analysis |
|---|---|---|---|---|
| Roman, 1997 (48) | Antibiotics/*27*<br>*Routinely collected health data* | Classification system not stated<br>Main outcomes: Leukemias/*143*, NHL/*34*<br>Secondary outcomes: ALL/*113*, AML/*15* | Case-control/*429*<br>*England, ages 3 months to 29 years*<br>Conditional logistic regression | Matching on hospital catchment area of birth, sex, year and month of birth |
| Ross, 2003 (19) | Amoxicillin/*34*, Ampicillin/*49*, Cefaclor/*7*, Cephalexin/*13*, Clotrimazole/*24*, Cotrimoxazole/*11*, Erythromycin/*43*, Metronidazole/*8*, Miconazole/*61* Nitrofurantoin/*12*, Nystatin/*11*, Penicillin V potassium/*11*, Triple sulfa/*10*, Rh(O) O immunoglobulin/*57*<br>*Routinely collected health data* | Classification system not stated<br>IL/*243*, ALL/*157*, AML/*77* | Case-control/*636*<br>*US, ages 0-18 months*<br>Conditional logistic regression | Matching on birth year and telephone area. Adjustment for maternal age, education and income |
| Salonen, 1976 (20) | Vaccination/*64*, Sulphonamides/*24*, Penicillins/*28*, Tetracyclines/*9*<br>*Routinely collected health data* | Classification: ICD<br>Main outcomes: Leukemias/*373*, brain tumors/*245*, other tumors/*354*<br>Secondary outcomes: Subgroups of the "other tumors"-category; kidney tumors/*96*, eye tumors/*37*, bone tumors/*56* (only reported for most commonly used medications) | Case-control/*1944*<br>*Finland, ages 0-14 years*<br>MacNemar's test, descriptive for rarely used medications | Matching on date of birth |
| Sanders and Draper, 1979 (49) | Isoniazid/*12*, Streptomycin/*9*<br>*Maternal retrospective report and medical records, findings from the two data sources are reported separately* | Classification system not stated<br>Any cancer/*27* | Case-controls/*40*<br>*UK, Oxford Survey of Childhood Cancers, ages 0-15. All children were born to mothers with pulmonary tuberculosis*<br>Descriptive statistics | It would appear that matching was done, but it is unclear |
| Shaw, 2004 (23) | Antibiotics/*87*<br>*Maternal retrospective report* | Classification: ICD<br>ALL/*789* | Case-control/*1578*<br>*Canada, age 0-14 years*<br>Conditional logistic regression | Matching on sex and age at diagnosis. Adjustment for maternal age and education |

| Reference | Exposure | Outcome | Materials and methods | |
|---|---|---|---|---|
| First author, year | Medication/*exposed*<br>*Ascertainment*<br>Exposure window: during pregnancy, unless otherwise stated | Classification system, type of cancer/*cases* | Design/*sample size*<br>*Study characteristics*<br>Statistical analysis | Method to account for confounding in the main analysis |
| Shaw, 2006 (50) | Antibiotics/*30*<br>*Maternal retrospective report* | Classification: ICD-O<br>Main outcome:<br>CNS tumor/*272*<br>Secondary outcomes:<br>Ependymoma/*27*, astrocytoma/*120*, PNET/*80* | Case-control/*544*<br>*Canada, ages 0-15 years*<br>Conditional logistic regression | Matching on sex and age at diagnosis. Adjustment for maternal education |
| Shu, 1995 (24) | Antibiotics/*78*<br>*Maternal retrospective report* | Classification system not stated<br>Malignant germ-cell tumors/*105* | Case-control/*744*<br>*US and Canada, CCG (Children's Cancer Group), ages 0-15 years*<br>Unconditional logistic regression | Matching on telephone area code. Adjustment for child age, sex, gestational age, number of siblings, maternal education, and smoking |
| Stålberg, 2010 (25) | Anti-infectives/*189*, antifungals/*105*, penicillins/*55*, antibiotics, others/*60*<br>*Routinely collected health data* | Classification: ICD<br>Brain tumor (ICD-7 code 193)/*512* | Case-control/*1037*<br>*Sweden, ages 0-15 years*<br>Logistic regression | Matching on sex and birth year. Adjustment for maternal age, parity, country of birth, and level of hospital (primary, secondary, or tertiary) |
| Thapa, 1998 (51) | Metronidazole/*79,716 person-years*<br>*Routinely collected health data, validated by using two different definitions of no exposure* | Classification: ICD<br>Main outcome: All cancer/*175*<br>Secondary outcomes:<br>leukemias/*42*, CNS tumors/*30*, neuroblastoma/*28*, other cancers/*75* | Cohort/*1,172,696 person-years*<br>*US, ages 0-5 years*<br>Poisson regression | Adjustment for maternal age, residence area, ethnicity, marital status, education, and parity |
| Van Duijn, 1994 (52) | Antibiotics in the year before or during pregnancy/*8*<br>*Maternal retrospective report, validated by comparison to associations in another case-control study where cases of a different type of cancer did not report more medication use than controls* | Classification system not stated<br>ANLL/*80* | Case-control/*320*<br>*The Netherlands, ages 0-14 years, study reported results for both population- and cancer controls*<br>Conditional logistic regression | Matching on sex and age. Adjustment for sex, year of birth, socioeconomic position, in pregnancy: alcohol use, smoking, occupational hydrocarbon exposure, other medication use, ultrasound, x-rays, viral infections |

| Reference | Exposure | Outcome | Materials and methods | |
|---|---|---|---|---|
| First author, year | Medication/*exposed*<br>*Ascertainment*<br>Exposure window: during pregnancy, unless otherwise stated | Classification system, type of cancer/*cases* | Design/*sample size*<br>*Study characteristics*<br>Statistical analysis | Method to account for confounding in the main analysis |
| Van Steensel-Moll, 1985 (53) | Antibiotics/*27*, Vaccination/*31*<br>*Maternal retrospective report* | Classification: ICD<br>Leukemias/*519* | Case-control/*1026*<br>*The Netherlands, ages 0-15 years*<br>Logistic regression | Matching on date of birth, sex and municipality. Adjustment for age and sex |
| Walsh, 2019 (54) | H1N1 influenza vaccination/*31,295*<br>*Routinely collected health data* | Classification: ICD<br>Neoplasm/*145* | Cohort/*104,249*<br>*Canada, ages 0-5 years*<br>Cox regression | Inverse probability of treatment weighting. The variables in the propensity score were maternal age, parity, smoking, season of conception, antenatal care provider, pre-existing comorbidity, obstetric complications, antenatal steroid use, income, residence area, public health unit region and multifetal gestation |
| Wen, 2002 (27) | Antibiotics in the year before pregnancy, through pregnancy and/or during breastfeeding/*400*<br>*Maternal retrospective report* | Classification system not stated<br>ALL/*1842* | Case-control/*3828*<br>*US, ages 0-14 years*<br>Conditional logistic regression | Matching on age, ethnicity, and telephone area.<br>Adjustment for parental income, education, ethnicity, smoking and alcohol use |
| Ye, 2019 (55) | Antibiotics (ATC class J01)/*98,997*<br>*Routinely collected health data* | Classification: ICCC<br>Main outcome: Any cancer (excluding nonmelanoma skin cancer)/*361*<br>Secondary outcomes: Leukemias/*141*, ALL/*114* | Cohort/*262,116*<br>*Canada, ages 0-19 years*<br>Cox regression | Adjustment for maternal age, income, prenatal use of tobacco, alcohol or illicit drugs |
| ANTINAUSEANTS | | | | |
| Buckley, 1989 (33) | Antinauseants in the year before pregnancy or during pregnancy/*19*<br>*Maternal retrospective report* | Classification system not stated<br>Hepatoblastoma/*75* | Case-control/*150*<br>*US and Canada, CCSG, upper age limit not stated, but the* | Matching on age |

| Reference<br>First author, year | Exposure<br>Medication/*exposed*<br>*Ascertainment*<br>Exposure window: during pregnancy, unless otherwise stated | Outcome<br>Classification system, type of cancer/*cases* | Materials and methods<br>Design/*sample size*<br>*Study characteristics*<br>Statistical analysis | <br>Method to account for confounding in the main analysis |
|---|---|---|---|---|
| | | | *majority of children were under 2 years of age*<br>Conditional logistic regression | |
| Bunin, 1989 (34) | Antinauseants/*45*<br>*Maternal retrospective report* | Classification system not stated<br>Nonheritable retinoblastoma/*115*<br>Sporadic heritable retinoblastoma (bilateral cases or unilateral cases with 13q chromosomal deletion)/*67*<br>Cases with family history were excluded | Case-control/*383*<br>*US and Canada, CCSG, upper age limit not stated*<br>Conditional logistic regression | Matching on telephone area code, year of birth and ethnicity. Adjustment for paternal education |
| Bunin, 1994 (2) | Antinauseants (specific to morning sickness)/*37*,<br>Antinauseants (other)/*11*<br>*Maternal retrospective report* | Classification system not stated<br>Astrocytoma/*155*<br>PNET/*166* | Case-control/*332*<br>*US and Canada, Children's Cancer Group, ages 0-5 years*<br>Conditional logistic regression | Matching on telephone area code, year of birth and ethnicity. Adjustment for income (only for astrocytoma) |
| Cardy, 2006 (35) | Antiemetics/*33*<br>*Maternal retrospective report* | Classification: ICD-O<br>Main outcome:<br>Any brain tumor/*1218*<br>Secondary outcomes:<br>Astroglial/*623*, PNET/*259*, other glial/*327* by morphological codes | Case-control/*3441*<br>*Europe, North America, Australia, SEARCH, upper age limit 19 years*<br>Unconditional logistic regression | Matching, not further specified. Adjustment for center, age, sex, year of birth, and parental education. Vitamin supplementation (unclear whether in mothers or children) was considered as both a confounder and effect modifier |
| Gilman, 1989 (6) | Antinauseants/*996*<br>*Maternal retrospective report and medical records. Children are considered exposed if medication use is present in either source* | Classification system not stated<br>Any cancer/*8059* | Case-control/*16,118*<br>*UK, OSCC, age not specified, only that cases are childhood deaths from cancer*<br>Conditional logistic regression | Matching on date of birth, sex and district. Adjustment for abdominal x-rays in pregnancy, birth order among live born siblings, maternal age, and socioeconomic position |

| Reference | Exposure | Outcome | Materials and methods | |
|---|---|---|---|---|
| First author, year | Medication/*exposed*<br>*Ascertainment*<br>Exposure window: during pregnancy, unless otherwise stated | Classification system, type of cancer/*cases* | Design/*sample size*<br>*Study characteristics*<br>Statistical analysis | Method to account for confounding in the main analysis |
| Hartley, 1988 (8) | Antiemetics/*not reported*<br>*Maternal retrospective report, validated in prescription data* | Classification system not stated<br>Bone or soft tissue sarcomas/*73* | Case-control/*219*<br>*UK, IRESCC, ages 0-15 years*<br>Mantel-Haenszel, 95% CI by Cornfield's formula | Matching. Matching factors not reported, but they refer to a paper where methods are described (age and sex) |
| Kramer, 1987 (11) | Antinauseants/*not reported*<br>*Maternal retrospective report, validated in medical records for a subsample. Validity compared between cases and controls* | Classification system not stated<br>Neuroblastoma/*104* | Case-control/*205*<br>*US, median age 1 year*<br>90% CI calculated by Miettinen's technique | Matching on telephone area code, ethnicity, and year of birth |
| Kuijten, 1990 (12) | Antinauseants/*54*<br>*Maternal retrospective report* | Classification system not stated<br>Astrocytoma/*163* | Case-control/*326*<br>*US, ages 0-14 years*<br>Conditional logistic regression | Matching on telephone area code, ethnicity, and age |
| Kwan, 2007 (44) | Antinauseants 3 months before and/or during pregnancy/*44*<br>*Maternal retrospective report* | Classification: ICD-O<br>Main outcome: Leukemia/*365*<br>Secondary outcome: ALL/*311* | Case-control/*825*<br>*US, NCCLS, ages 0-14 years*<br>Conditional logistic regression | Matching on date of birth, sex, ethnicity (of child and mother), and for part of the recruitment also county of residence at birth. Adjustment for maternal age, income, and education |
| McKinney, 1985 (56) | Debendox/*92*, Other antiemetics/*22*<br>Also analysis by duration<br>*Maternal retrospective report and medical records, findings from the two data sources are reported separately* | Classification system not stated<br>Main outcome: Any cancer/*555*<br>Secondary outcomes: Leukemias or lymphomas/*245*<br>Other cancers/*310* | Case-control/*1665*<br>*UK, IRESCC, ages 0-15 years*<br>Maximum likelihood, not specified further | Matching on age and sex |
| McKinney, 1999 (15) | Antinauseants/*14*<br>*Routinely collected health data* | Classification: ICCC<br>Main outcome: Leukemia/*144*<br>Secondary outcome: ALL/*124*, lymphomas/*45*, CNS tumors/*75*, other solid tumors/*126* | Case-control/*415*<br>*Scotland, ages 0-14 years*<br>Conditional logistic regression | Matching on age, area of residence, and sex |
| Michalek, 1996 (29) | Antinauseants/*46*<br>*Maternal retrospective report* | Classification system not stated<br>Neuroblastoma/*183* | Case-control/*555*<br>*US, ages 0-14 years*<br>Unconditional logistic regression | Matching on year of birth |

| Reference | Exposure | Outcome | Materials and methods | |
|---|---|---|---|---|
| First author, year | Medication/*exposed*<br>*Ascertainment*<br>Exposure window: during pregnancy, unless otherwise stated | Classification system, type of cancer/*cases* | Design/*sample size*<br>*Study characteristics*<br>Statistical analysis | Method to account for confounding in the main analysis |
| Pombo de Oliveira, 2006 (17) | Antiemetic 3 months before pregnancy, during pregnancy, or during breastfeeding/*43*<br>*Maternal retrospective report* | Classification system not stated<br>Infant acute leukemias/*202* | Case-control/*642*<br>*Brazil, ages 0-21 months*<br>Unconditional logistic regression | Matching on age and region. Adjustment for region, sex, birth weight, maternal age, and income |
| Ross, 2003 (19) | Bendectin (Dicyclomine, Doxylamine, Pyridoxine)/*12*<br>Prochlorperazine/*9*<br>Trimethobenzamide HCL/*7*<br>*Routinely collected health data* | Classification system not stated<br>IL/*243*, ALL/*157*, AML/*77* | Case-control/*636*<br>*US, ages 0-18 months*<br>Conditional logistic regression | Matching on birth year and telephone area. Adjustment for maternal age, education and income |
| Robison, 1989 (37) | Bendectin or other morning sickness tablets in the year before pregnancy, through pregnancy and/or during breastfeeding/*71*<br>Also analyses by duration and frequency<br>*Maternal retrospective report, validated by comparisons to associations in four other independent case-control studies where cases of different cancers did not report more medication use than controls* | Classification system not stated<br>ANLL/*204* | Case-control/*408*<br>*US, ages 0-18 years*<br>Conditional logistic regression | Matching on date of birth, ethnicity, and telephone area code, stated that they adjust for confounders, but these are not specified |
| Schüz, 2007 (21) | Antinauseants/antiemetics/*113*<br>*Maternal retrospective report from questionnaire data validated through phone interview* | Classification system not stated<br>ALL/*650*, AML/*105*, NHL/*172*, CNS tumors/*399*, neuroblastoma/*157*, nephroblastoma/*147*, bone tumor/*97*, soft tissue sarcoma/*137* | Case-control/*2652*<br>*Germany, ages 0-14 years*<br>Frequency matched conditional logistic regression | Matching on sex, date of birth, and district. Adjustment for age, sex, year of birth, degree of urbanization, and socioeconomic position |
| Sharpe & Franco, 1996 (22) | Metoclopramide/*50*, Antiemetics/*86*<br>*Maternal retrospective report* | Classification system not stated<br>Nephroblastoma/*109* | Case-control/*327*<br>*Brazil, age range not reported, mean age 3 years and 5 months*<br>Logistic regression | Matching on age, sex, and interviewer, stratified on socioeconomic position |
| Shaw, 2004 (23) | Anti-nausea medication/*115*<br>*Maternal retrospective report* | Classification: ICD<br>ALL/*789* | Case-control/*1578*<br>*Canada, age 0-14 years*<br>Conditional logistic regression | Matching on sex and age at diagnosis. Adjustment for maternal age and education |

| Reference<br>First author, year | Exposure<br>Medication/*exposed*<br>*Ascertainment*<br>Exposure window: during pregnancy, unless otherwise stated | Outcome<br>Classification system, type of cancer/*cases* | Materials and methods<br>Design/*sample size*<br>*Study characteristics*<br>Statistical analysis | <br>Method to account for confounding in the main analysis |
|---|---|---|---|---|
| Shu, 1995 (24) | Medicine for nausea or vomiting/*151*<br>*Maternal retrospective report* | Classification system not stated<br>Malignant germ-cell tumors/*105* | Case-control/*744*<br>*US and Canada, CCG (Children's Cancer Group), ages 0-15 years*<br>Unconditional logistic regression | Matching on telephone area code. Adjustment for child age, sex, gestational age, number of siblings, maternal education, and smoking |
| Stålberg, 2010 (25) | Antiemetics/*106*<br>*Routinely collected health data* | Classification: ICD<br>Brain tumor (ICD-7 code 193)/*512* | Case-control/*1037*<br>*Sweden, ages 0-15 years*<br>Logistic regression | Matching on sex and birth year. Adjustment for maternal age, parity, country of birth, and level of hospital (primary, secondary, or tertiary) |
| Wen, 2002 (27) | Bendectin or other morning sickness tablets in the year before pregnancy, through pregnancy and/or during breastfeeding/*231*<br>*Maternal retrospective report* | Classification system not stated<br>ALL/*1842* | Case-control/*3828*<br>*US, ages 0-14 years*<br>Conditional logistic regression | Matching on age, ethnicity, and telephone area. Adjustment for parental income, education, ethnicity, smoking and alcohol use |
| ANTITUSSIVES/ANTI-ASTHMATICS | | | | |
| Bonaventure, 2015 (1) | Respiratory system (ATC group R)/*409*<br>*Routinely collected health data* | Classification: ICD<br>ALL/*725*, AML/*128*, NHL/*83*, HL/*31*, astrocytoma/*100*, medulloblastoma/*48*, rhabdomyosarcoma/*54*, Ewings sarcoma/*20*, neuroblastoma/*78*, nephroblasoma/*60* | Case-control/*4122*<br>*UK, UKCCS, mean age 5.8 (max age 15 years)*<br>Unconditional logistic regression | Matching on sex, month and year of birth, and region of residence. Adjustment for year of birth |
| Buckley, 1989 (33) | Cold or cough medicines/*12*<br>*Maternal retrospective report* | Classification system not stated<br>Hepatoblastoma/*75* | Case-control/*150*<br>*US and Canada, CCSG, upper age limit not stated, but the majority of children were under 2 years of age*<br>Conditional logistic regression | Matching on age |
| Carozza, 1995 (3) | Epinephrine/*1*<br>*Maternal retrospective report* | Classification: Histological<br>Typing of Tumors of the Central | Case-control/*1444* | Matching on age, sex and maternal ethnicity |

| Reference<br>First author, year | Exposure<br>Medication/*exposed*<br>*Ascertainment*<br>Exposure window: during pregnancy, unless otherwise stated | Outcome<br>Classification system, type of cancer/*cases* | Materials and methods<br>Design/*sample size*<br>*Study characteristics*<br>Statistical analysis | <br>Method to account for confounding in the main analysis |
|---|---|---|---|---|
| | | Nervous System. International Histological Classification of Tumors<br>Any brain tumors/*361* | *US, SEER, upper age limit 18 years*<br>Conditional logistic regression, for main exposure (any n-nitrosatable drugs), descriptive statistics for medications stated here | |
| Cook, 2004 (4) | Albuterol/*8*, Phenylephrine/*14*, Phenylpropanolamine/*25*, Pseudoephedrine/*116*, Terbutaline/*12*, Dextromethorphan/*14*, Guaifenesin/*68*<br>Each of the above during pregnancy or breastfeeding<br>*Maternal retrospective report* | Classification system not stated<br>Neuroblastoma/*504* | Case-control/*1008*<br>*US and Canada, Children's Cancer Group, upper age limit 18 years*<br>Conditional logistic regression | Matching on age. Adjustment for maternal age, education, and ethnicity |
| Gilman, 1989 (6) | Expectorants/*54*, Bronchospasm relaxants/*78*<br>*Maternal retrospective report and medical records. Children are considered exposed if medication use is present in either source* | Classification system not stated<br>Any cancer/*8059* | Case-control/*16,118*<br>*UK, OSCC, age not specified, only that cases are childhood deaths from cancer*<br>Conditional logistic regression | Matching on date of birth, sex and district. Adjustment for abdominal x-rays in pregnancy, birth order among live born siblings, maternal age, and socioeconomic position |
| McCredie, 1994 (13) | Cold or cough remedies in the month before or during pregnancy/*7*<br>*Maternal retrospective report* | Classification: ICD<br>Brain or cranial nerves tumors/*82* | Case-control/*246*<br>*Australia, ages 0-14 years*<br>Conditional logistic regression | Matching on sex and age. Adjustment for parental education |
| Ross, 1996 (57) | Cold medications/*77*<br>*Maternal retrospective report* | Classification system not stated<br>IL/*84* | Case-control/*181*<br>*US, ages 0-1 year*<br>Conditional logistic regression | Matching on year of birth, geography and ethnicity. Adjustment for maternal education |
| Ross, 2003 (19) | Terbutaline/*15*, Theophylline salts/*6*<br>*Routinely collected health data* | Classification system not stated<br>IL/*243*, ALL/*157*, AML/*77* | Case-control/*636*<br>*US, ages 0-18 months*<br>Conditional logistic regression | Matching on birth year and telephone area. Adjustment for maternal age, education and income |

| Reference | Exposure | Outcome | Materials and methods | |
|---|---|---|---|---|
| First author, year | Medication/*exposed*<br>*Ascertainment*<br>Exposure window: during pregnancy, unless otherwise stated | Classification system, type of cancer/*cases* | Design/*sample size*<br>*Study characteristics*<br>Statistical analysis | Method to account for confounding in the main analysis |
| Robison, 1989 (37) | Cold/cough medication in the year before pregnancy, through pregnancy and/or during breastfeeding/*61*<br>Also analyses by duration and frequency<br>*Maternal retrospective report, validated by comparisons to associations in four other independent case-control studies where cases of different cancers did not report more medication use than controls* | Classification system not stated<br>ANLL/*204* | Case-control/*408*<br>*US, ages 0-18 years*<br>Conditional logistic regression | Matching on date of birth, ethnicity, and telephone area code, stated that they adjust for confounders, but these are not specified |
| Salonen, 1976 (20) | Parasympatholytics/*20*<br>*Routinely collected health data* | Classification: ICD<br>Main outcomes: Leukemias/*373*, brain tumors/*245*, other tumors/*354*<br>Secondary outcomes: Subgroups of the "other tumors"-category; kidney tumors/*96*, eye tumors/*37*, bone tumors/*56* (only reported for most commonly used medications) | Case-control/*1944*<br>*Finland, ages 0-14 years*<br>MacNemar's test, descriptive for rarely used medications | Matching on date of birth |
| Schüz, 2007 (21) | Cold medications/*282*<br>*Maternal retrospective report from questionnaire data validated through phone interview* | Classification system not stated<br>ALL/*650*, AML/*105*, NHL/*172*, CNS tumors/*399*, neuroblastoma/*157*, nephroblastoma/*147*, bone tumor/*97*, soft tissue sarcoma/*137* | Case-control/*2652*<br>*Germany, ages 0-14 years*<br>Frequency matched conditional logistic regression | Matching on sex, date of birth, and district. Adjustment for age, sex, year of birth, degree of urbanization, and socioeconomic position |
| Shaw, 2004 (23) | Anti-asthmatics/*16*<br>*Maternal retrospective report* | Classification: ICD<br>ALL/*789* | Case-control/*1578*<br>*Canada, age 0-14 years*<br>Conditional logistic regression | Matching on sex and age at diagnosis. Adjustment for maternal age and education |
| Stålberg, 2010 (25) | Anti-asthmatics/*68*<br>*Routinely collected health data* | Classification: ICD<br>Brain tumor (ICD-7 code 193)/*512* | Case-control/*1037*<br>*Sweden, ages 0-15 years*<br>Logistic regression | Matching on sex and birth year. Adjustment for maternal age, parity, country of birth, and level of hospital (primary, secondary, or tertiary) |

| Reference | Exposure | Outcome | Materials and methods | |
|---|---|---|---|---|
| First author, year | Medication/*exposed*<br>*Ascertainment*<br>Exposure window: during pregnancy, unless otherwise stated | Classification system, type of cancer/*cases* | Design/*sample size*<br>*Study characteristics*<br>Statistical analysis | Method to account for confounding in the main analysis |
| Tegethoff, 2013 (58) | Asthma medications/*2510*<br>*Maternal prospective report* | Classification: ICD<br>Neoplasms/*721* | Cohort/*66,712*<br>*Denmark, ages 3-9 years*<br>Cox regression | Adjustment for maternal age, socioeconomic position, parity, general health, smoking, and child sex |
| Wen, 2002 (27) | Prescription cold or cough remedies in the year before pregnancy, through pregnancy and/or during breastfeeding/*95*<br>*Maternal retrospective report* | Classification system not stated<br>ALL/*1842* | Case-control/*3828*<br>*US, ages 0-14 years*<br>Conditional logistic regression | Matching on age, ethnicity, and telephone area.<br>Adjustment for parental income, education, ethnicity, smoking and alcohol use |
| FERTILITY MEDICATIONS | | | | |
| Ajrouche, 2014 (59) | Any fertility treatment/*139*, Clomiphene citrate/*36*, Gonadotropins/*63*, Dopamine agonists/*6*<br>*Maternal retrospective report, validated in a sensitivity analysis assuming worst case differential misclassification* | Classification system not stated<br>Main outcome: Acute leukemias/*747*<br>Secondary outcomes: ALL/*636*, AML/*100* | Case-control/*2168*<br>*France, ESCALE, ages 0-15 years*<br>Unconditional logistic regression | Matching on age and sex.<br>Adjustment for age, sex, maternal age, birth order, maternal education or occupation |
| Bradbury & Jick, 2004 (60) | IVF/*176*<br>*Routinely collected health data* | Classification system not stated<br>Retinoblastoma/*24* | Cohort/*358,270*<br>*UK, age 0-5 years*<br>Descriptive statistics | No adjustment |
| Brinton, 2004 (61) | Any ovarian stimulation/*349*,<br>Clomiphene citrate/*181*, hCG/*184*, hMG/*49*<br>Each of the above ever before birth of index child. Also analysis by number of treatments.<br>Comparators are women with infertility. Some may be exposed to other fertility drugs<br>*Routinely collected health data* | Classification system not stated<br>Main outcome: Any cancer/*47*<br>Secondary outcomes:<br>Hematopoietic tumors (ALL, AML, HL, NHL)/*19*<br>Neural tumors (neuroblastomas, astrocytomas, medulloblastomas and other brain tumors)/*19*<br>Other/*9* | Case-cohort/915<br>*Denmark, ages 0-20 years*<br>Cox regression | Matching on maternal age and calendar year at entry into infertility cohort. Adjustment for maternal age at birth |
| Foix-L'Hélias, 2012 (62) | Any infertility treatment/*1575*<br>Ovarian stimulation/*837*<br>*Maternal retrospective report, validated in a sensitivity analysis assuming worst case differential misclassification* | Classification system not stated<br>Non-familial retinoblastoma/*244*, for the exposure "any fertility treatment" also by mutation | Case-cohort/*28,414*<br>*France, ages 0-5 years*<br>Logistic regression for main exposure, descriptive statistics for the exposure reported here | No adjustment in the analysis on medication use |

| Reference<br>First author, year | Exposure<br>Medication/*exposed*<br>*Ascertainment*<br>Exposure window: during pregnancy, unless otherwise stated | Outcome<br>Classification system, type of cancer/*cases* | Materials and methods<br>Design/*sample size*<br>*Study characteristics*<br>Statistical analysis | <br>Method to account for confounding in the main analysis |
|---|---|---|---|---|
| | | status and by multifocality (yes/no) | | |
| Hargreave, 2015 (63) | Any fertility medication/*794*,<br>Clomiphene citrate/*443*, Gonadotropins/*375*, GnRH/*272*, hCG/*604*, Progesterone/*260*<br>Each of the above ever before birth of index child<br>Also analysis by number of treatments<br>Comparators were diagnosed with infertility<br>*Routinely collected health data* | Classification: ICCC<br>Main outcome:<br>Any cancer/*129*<br>Secondary outcomes:<br>Leukemias/*48*, lymphomas/*10*, SNS tumors/*13*, CNS tumors/*33*, other cancers/*25* | Case-cohort/*1418*<br>*Denmark, ages 0-19 years*<br>Cox regression | Matching on maternal age and calendar year at entry into infertility cohort |
| Hargreave, 2019 (64) | Any fertility medication/*89,334*,<br>Clomiphene citrate/*33,835*, Gonadotropins/*57,136*, hCG/*68,181*, GnRH analogues/*38,653*, Progesterone/*41,628*, Estrogen/*16,948*<br>Uses disease comparators in a sensitivity analysis.<br>*Routinely collected health data* | Classification system not stated<br>Main outcome:<br>Any cancer/*2217*<br>Secondary outcomes:<br>Leukemias/*648*, lymphomas/*242*, SNS tumors/*130*, CNS tumors/*541*, other cancers/*661* | Cohort/*1,085,172*<br>*Denmark, ages 0-19, mean 11.3*<br>Cox regression | Adjustment for year of birth |
| Heck, 2012 (65) | Fertility enhancing drugs/*not reported*<br>*Routinely collected health data* | Classification system not stated<br>Retinoblastoma/*609* | Case-control/*209,660*<br>*US, APCC, ages 0-5 years*<br>Logistic regression | Matching on year of birth.<br>Adjustment for year of birth, paternal age, county of residence, and maternal place of birth |
| Heck, 2013 (66) | Fertility treatment/*not reported*<br>*Routinely collected health data* | Classification system not stated<br>Hepatoblastoma/*261* | Case-control/*218,538*<br>*California, ages 0-5 years*<br>Unconditional logistic regression | Matching on year of birth.<br>Adjustment for year of birth, maternal age, and maternal ethnicity |
| Heck, 2015 (9) | Fertility medication/*28*<br>*Maternal retrospective report* | Classification system not stated<br>Sporadic retinoblastoma, unilateral/*187*<br>Sporadic retinoblastoma, bilateral/*95* | Case-control/*426*<br>*US/Canada, ages 0-15 years*<br>Unconditional logistic regression | Matching on year of birth.<br>Adjustment for maternal ethnicity, education, income, maternal age, and smoking in pregnancy |
| Källen, 2009 (67) | IVF/*26692*<br>*Routinely collected health data* | Classification system not stated<br>Any cancer/*6458* | Cohort/*2,425,733*<br>*Sweden, ages 0-23 years* | Adjustment for year of birth |

| Reference | Exposure | Outcome | Materials and methods | |
|---|---|---|---|---|
| First author, year | Medication/*exposed*<br>*Ascertainment*<br>Exposure window: during pregnancy, unless otherwise stated | Classification system, type of cancer/*cases* | Design/*sample size*<br>*Study characteristics*<br>Statistical analysis | Method to account for confounding in the main analysis |
| | | Lists the specific types of cancer observed in exposed children | Mantel-Haenszel, CI calculated by Miettinen's technique | |
| Klip, 2001 (68) | Any assisted reproduction or fertility drugs/*9479*<br>Comparators were women with subfertility diagnosis who conceived spontaneously<br>*Routinely collected health data* | Classification system not stated<br>Any cancer/*16* | Cohort/*17,000*<br>*The Netherlands (OMEGA), ages 0-15 years*<br>Cox regression | Adjustment for sex |
| Lerner-Geva, 2017 (69) | IVF/ICSI/*9042*<br>*Routinely collected health data* | Classification: ICCC<br>Main outcome: Any cancer/*382*<br>Secondary outcomes: Cancers by ICCC-3 groups: Leukemias, myeloproliferative diseases, and myelodysplastic diseases/*94*, lymphomas and reticuloendothelial neoplasms/*67*, CNS and miscellaneous intracranial and intraspinal neoplasms/*72*, neuroblastoma and other peripheral nervous cell tumors/*58*, retinoblastoma/*11*, renal tumors/*27*, hepatic tumors/*2*, malignant bone tumors/*10*, soft tissue and other extraosseous sarcomas/*27*, germ cell tumors, trophoblastic tumors, and neoplasms of gonads/*2*, other malignant epithelial neoplasms and malignant melanomas/*9*, other and unspecified malignant neoplasms/*3* | Cohort/*220,805*<br>*Israel, ages 0-12 years*<br>Poisson regression | Adjustment for maternal age, education, ethnicity, child plurality, sex, birth weight, and congenital malformations |
| Mallol-Mesnard, 2008 (70) | Assisted reproduction/*93*<br>*Maternal retrospective report* | Classification: ICCC<br>Main outcome: Any CNS tumor/*209* | Case-control/*1890*<br>*France, ESCALE, ages 0-14 years* | Matching on age, sex, and number of siblings.<br>Adjustment for age and sex |

| Reference<br>First author, year | Exposure<br>Medication/*exposed*<br>*Ascertainment*<br>Exposure window: during pregnancy, unless otherwise stated | Outcome<br>Classification system, type of cancer/*cases* | Materials and methods<br>Design/*sample size*<br>*Study characteristics*<br>Statistical analysis | Method to account for confounding in the main analysis |
|---|---|---|---|---|
| | | Secondary outcomes: Ependymomas/*33*, embryonal tumors/*100*, astrocytomas/*26*, other gliomas/*45* | Unconditional logistic regression | |
| McCredie, 1994 (13) | Fertility drugs in the month before or during pregnancy/*5*<br>*Maternal retrospective report* | Classification: ICD<br>Brain or cranial nerves tumors/*82* | Case-control/*246*<br>*Australia, ages 0-14 years*<br>Conditional logistic regression | Matching on sex and age.<br>Adjustment for parental education |
| McLaughlin, 2006 (71) | Presumptive fertility treatment (IVF, other fertility treatment, or triplet pregnancy)/*28*<br>*Routinely collected health data* | Classification system not stated<br>Hepatoblastoma/*58* | Case-control/*6114*<br>*US ages 0-5 years*<br>Not specified | Matching on birth year.<br>Adjustment for birth year and birth weight |
| Munzer, 2008 (72) | Fertility treatment/*91*, Ovarian stimulation only/*46*<br>*Maternal retrospective report, validated in a sensitivity analysis restricted to women who had access to their children's health card during interview* | Classification: ICCC<br>Neuroblastoma/*191*, also neuroblastoma subgroups by MYCN oncogene amplification status | Case-control/*1872*<br>*France, ESCALE, 0-14 years*<br>Unconditional logistic regression | Matching on age and sex.<br>Adjustment for matching factors |
| Olshan, 1999 (73) | Any fertility drug/*54*, Any ovulation-inducing drug/*49*, Clomiphene/*46*<br>Each of the above ever, in the 2-12 months prior to pregnancy, or 1 month prior to pregnancy<br>*Maternal retrospective report* | Classification system not stated<br>Neuroblastoma/*504*, also neuroblastoma subgroups by MYCN oncogene amplification status in an exploratory analysis | Case-control/*1008*<br>*US and Canada, ages 0-19 years*<br>Conditional logistic regression | Matching on date of birth.<br>Adjustment for maternal ethnicity, education, and income |
| Petridou, 2012 (74) | IVF/*137*<br>*Maternal retrospective report, validated from medical record in Greece, routinely collected health data in Sweden* | Classification: ICD<br>Main outcome: Leukemias/*1334*<br>Secondary outcomes: ALL/*1095*, lymphoma/*348* | Case-control *in Greece, ages 0-14, pooled with* case-cohort *in Sweden, ages 0-12 years/8683*<br>Not specified how OR with 95% CI were estimated, but results from the two countries were combined using a fixed-effects model | Matching on age and sex.<br>Adjustment for birth weight, birth order, maternal age, education, smoking during pregnancy, and plurality |
| Puumala, 2007 (75) | Infertility medication, ever/*32*<br>*Maternal retrospective report* | Classification system not stated<br>Main outcome: Leukemias/*158*<br>Secondary outcomes: ALL/*97*, AML/*61* | Case-control/*331*<br>*US, Children's Oncology Group, ages 0-19 years, all* | Matching on age. Adjustment for maternal age, ethnicity, education, and child sex |

| Reference<br>First author, year | Exposure<br>Medication/*exposed*<br>*Ascertainment*<br>Exposure window: during pregnancy, unless otherwise stated | Outcome<br>Classification system, type of cancer/*cases* | Materials and methods<br>Design/*sample size*<br>*Study characteristics*<br>Statistical analysis | <br>Method to account for confounding in the main analysis |
|---|---|---|---|---|
| | | | *cases and controls have Down syndrome.*<br>Unconditional logistic regression | |
| Puumala, 2010 (76) | Ovarian stimulating drugs, unclear whether ever or for index pregnancy/*34*<br>*Maternal retrospective report* | Classification system not stated<br>Main outcome: Infant leukemias/*443*<br>Secondary outcomes: ALL/*264*, AML/*172*, also analyzed by MLL gene translocation status | Case-control/*767*<br>*US, Children's Oncology Group, ages 0-1 years*<br>Logistic regression | Matching on year of birth and region of residence.<br>Adjustment for matching factors, maternal age, education, ethnicity, smoking during pregnancy, income, gestational age and birth weight |
| Puumala, 2011 (77) | Ovarian stimulating drugs in the year prior to or during index pregnancy/*16*<br>*Maternal retrospective report* | Classification system not stated<br>Main outcome: Germ cell tumors/*278*<br>Secondary outcomes: Gonadal germ cell tumors/*144*, non-gonadal germ cell tumors/*120* | Case-control/*700*<br>*US, Children's Oncology Group, ages 0-14 years*<br>Unconditional logistic regression | Matching on sex, age and telephone area code.<br>Adjustment for age, sex, gestational age, maternal age, ethnicity, education, and income |
| Puumala, 2012 (78) | Infertility medication, unclear whether ever or for index pregnancy/*79*<br>*Maternal retrospective report* | Classification system not stated<br>Hepatoblastoma/*383* | Case-control/*770*<br>*US, Children's Oncology Group, ages 0-5 years*<br>Logistic regression | Matching on birth weight, sex, year of birth, and region.<br>Adjustment for matching factors, maternal age, education, ethnicity, birth plurality, and gestational age |
| Reigstad, 2016 (79) | Assisted reproduction/*25,782*<br>*Routinely collected health data* | Classification: ICD<br>Main outcome: Any cancer/*4554*<br>Secondary outcomes: Main ICCC-3 diagnostic groups, and subgroups for leukemias, lymphomas and CNS tumors: ALL/*777*, AML/*178*, other leukemias/*74*, HL/*261*, NHL/*148*, astrocytoma/*368*, embryonal | Cohort/*1,628,658*<br>*Norway, ages 0-28 years, median follow-up 6.9 years in exposed, 13.7 years in unexposed*<br>Cox regression | Adjusted for calendar year at follow-up, region, maternal age, and birth order |

| Reference<br>First author, year | Exposure<br>Medication/*exposed*<br>*Ascertainment*<br>Exposure window: during pregnancy, unless otherwise stated | Outcome<br>Classification system, type of cancer/*cases* | Materials and methods<br>Design/*sample size*<br>*Study characteristics*<br>Statistical analysis | <br>Method to account for confounding in the main analysis |
|---|---|---|---|---|
| | | CNS tumors/*179*, other gliomas/*105*, other CNS tumors/*133*, neuroblastoma/*188*, retinoblastoma/*91*, renal/*254*, hepatic/*110*, soft tissue/*312*, other/*491* | | |
| Ross, 2003 (19) | Clomiphene/*10*<br>*Routinely collected health data* | Classification system not stated<br>IL/*243*, ALL/*157*, AML/*77* | Case-control/*636*<br>*US, ages 0-18 months*<br>Conditional logistic regression | Matching on birth year and telephone area. Adjustment for maternal age, education and income |
| Rudant, 2013 (80) | Ovulation induction drugs (without in vitro fertilization or artificial insemination)/*79*<br>*Maternal retrospective report* | Classification system not stated<br>Main outcome: Acute leukemias/*764*<br>Secondary outcomes: ALL/*648*, AML/*101* | Case-control /*2445*<br>*France, ages 0-15 years*<br>Unconditional logistic regression | Matching on sex and age. Adjustment for sex, age, parental occupation, and maternal age |
| Schüz, 1999 (81) | Hormonal treatment for infertility, unclear whether ever or for index pregnancy/*225*<br>*Maternal retrospective report from questionnaire data validated through phone interview* | Classification system not stated<br>Main outcomes: Acute leukemias/*1184*, NHL/*234*, CNS tumors/*399*, neuroblastomas/*160*, nephroblastomas/*147*, bone tumors/*97*, soft tissue sarcomas/*137*<br>Secondary outcomes: ALL/*1037*, common-ALL/*686*, pre-B-ALL/*121*, T-ALL/*99*, ANLL/*147* | Case-control/*4946*<br>*Germany, ages 0-14 years*<br>Frequency matched conditional logistic regression | Matching on sex, date of birth, and district. Adjustment for age, sex, year of birth, vicinity to nuclear installation, degree of urbanization, and socioeconomic position |
| Schüz, 2001 (82) | Hormonal treatment for infertility, unclear whether ever or for index pregnancy/*90*<br>*Maternal retrospective report from questionnaire data validated through phone interview* | Classification: International Neuroblastoma Staging System<br>Neuroblastoma/*166* | Case-control/*1866*<br>*Germany, ages 0-7 years.*<br>Frequency matched conditional logistic regression | Matching on sex, date of birth, and district. Adjustment for age, sex, year of birth, degree of urbanization, and socioeconomic position |
| Spector, 2019 (83) | IVF/*275,686*<br>*Routinely collected health data* | Classification: ICCC<br>Main outcome: Any cancer/*2363* | Cohort/*2,542,533*<br>*US, ages 0-10 years* | Adjustment for sex, plurality, maternal education, maternal |

| Reference | Exposure | Outcome | Materials and methods | |
|---|---|---|---|---|
| First author, year | Medication/*exposed*<br>*Ascertainment*<br>Exposure window: during pregnancy, unless otherwise stated | Classification system, type of cancer/*cases* | Design/*sample size*<br>*Study characteristics*<br>Statistical analysis | Method to account for confounding in the main analysis |
| | | Secondary outcomes: ICCC3 subgroup if more than 10 exposed cases, otherwise ICCC3 group: ALL/*606*, AML/*121*, lymphomas/*161*, astrocytomas/*231*, ependymomas/*53*, intracranial embryonal tumors/*96*, neuroblastomas/*307*, retinoblastomas/*141*, renal/*214*, hepatic/*83*, soft tissue sarcomas/*115*, germ cell/*54*, embryonal/*877* | Cox regression | ethnicity, maternal age, state of birth |
| Sundh, 2014 (84) | Assisted reproductive technology including IVF, ICSI and frozen embryo transfer/*91,796*<br>*Routinely collected health data* | Classification: ICCC, based on ICD-codes.<br>Main outcome: Any cancer/*819*<br>Secondary outcomes: ICCC-3 groups: Leukemias/*278*, lymphomas/*52*, CNS tumors/*156*, neuroblastomas and peripheral nerve cell tumors/*48*, retinoblastomas/*35*, renal tumors/*51*, hepatic tumors/*10*, malignant bone tumors/*35*, soft tissue/extraosseous sarcomas/*37*, germ cell tumors/*21*, other malignant epithelial neoplasms/*40*, other/not specified malignant neoplasms/*56* | Cohort/*450,215*<br>*Nordic countries (Sweden, Denmark, Finland and Norway),*<br>*mean length of follow up 9.5 years for all children*<br>Cox regression | Adjustment for country, maternal age, parity, sex, gestational age, birth defects and chromosomal aberrations |
| Wainstock, 2017 (85) | Fertility treatments: IVF/*2603*, ovulation induction/*1172*<br>*Maternal prospective report, validated in medical records* | Classification: ICD<br>Any malignant neoplasm/*1498* | Cohort/*242,187*<br>*Israel, children born between 1991 and 2013* | Adjusted for maternal age, birth weight, preterm birth, pregnancy-related hypertensive disorders, pre- |

| Reference<br>First author, year | Exposure<br>Medication/*exposed*<br>*Ascertainment*<br>Exposure window: during pregnancy, unless otherwise stated | Outcome<br>Classification system, type of cancer/*cases* | Materials and methods<br>Design/*sample size*<br>*Study characteristics*<br>Statistical analysis | <br>Method to account for confounding in the main analysis |
|---|---|---|---|---|
| | | | Kaplan-Meier survival curves and Cox-Mantel log rang test. | gestational diabetes, gestational diabetes |
| HIV-MEDICATIONS | | | | |
| Benhammou, 2008 (86) | Zidovudine and lamivudine/*4752*<br>Didanosine and other non-lamivudine molecules/*874*, Didanosine and lamivudine/*365*, Other combinations/*715*<br>Each of the above compared to zidovudine monotherapy<br>*Routinely collected health data* | Classification system not stated<br>Any cancer/*10* | Cohort/*8853*<br>*France, Enquête Périnatale Française, ages 0-5 years*<br>Cox regression | Adjustment for prematurity |
| Hleyhel, 2016 (87) | Zidovudine and lamivudine/*7533*<br>Abacavir/tenofovir and emtricitabine/lamivudine/*3239*,Stavudine/zalcitabine alone or in combination with other molecules except didanosine/*649*, Didanosine alone or in combination with other molecules/*1461*<br>Each of the above compared to zidovudine monotherapy<br>Also analysis by duration<br>*Routinely collected health data* | Classification system not stated<br>Any cancer/*21* | Cohort/*15,163*<br>*France, Enquête Périnatale Française, 0-15 years*<br>Cox regression | Adjustment for prematurity, peripartum zidovudine infusion and postnatal HIV medication to the child |
| IMMUNOSUPPRESSANTS/ANTINEOPLASTIC AGENTS | | | | |
| Chambers, 2019 (88) | Adalimumab /*257*<br>Both disease and population comparators<br>*Maternal prospective report* | Classification system not stated<br>Any cancer/*0* | Cohort/*602*<br>*US and Canada, MotherToBaby/OTIS, age 0-1 year*<br>Descriptive statistics for the cancer outcome | No adjustment for this outcome |
| Chaparro, 2018 (89) | Anti-TNFα/*388*<br>Disease comparator<br>*Routinely collected health data* | Classification system not stated<br>Any cancer/*0* | Cohort/*841*<br>*Europe, TEDDY, ages 0-15 years*<br>Descriptive statistics for the cancer outcome | No adjustment for this outcome |

| Reference | Exposure | Outcome | Materials and methods | |
|---|---|---|---|---|
| First author, year | Medication/*exposed*<br>*Ascertainment*<br>Exposure window: during pregnancy, unless otherwise stated | Classification system, type of cancer/*cases* | Design/*sample size*<br>*Study characteristics*<br>Statistical analysis | Method to account for confounding in the main analysis |
| Green, 1997 (90) | Cancer treatment any time before pregnancy/*92*<br>Comparators are children born to untreated wives of treated men.<br>*Routinely collected health data* | Classification system not stated<br>Any cancer/*0* | Cohort/*153*<br>*US, ages not reported*<br>Descriptive statistics for the comparison we report here | No adjustment for this outcome |
| Li, 1979 (91) | Cancer treatment any time before pregnancy/*159*<br>Comparators are cousins of exposed<br>*Routinely collected health data* | Classification system not stated<br>Any cancer/*2* | Cohort/*571*<br>*US, ages 0-10 years*<br>Descriptive statistics | No adjustment for this outcome |
| Ognjanovic, 2009 (30) | Steroids or immunosuppressants/*15*<br>*Maternal retrospective report* | Classification system not stated<br>Main outcome: Leukemias/*158*<br>Secondary outcomes: ALL/*97*, AML/*61* | Case-control/*331*<br>*US, Children's Oncology Group, ages 0-19 years, all cases and controls have Down syndrome*<br>Descriptive statistics | Matching on age and sex |
| Ross, 2003 (19) | Hydrocortisone/*21*<br>*Routinely collected health data* | Classification system not stated<br>IL/*243*, ALL/*157*, AML/*77* | Case-control/*636*<br>*US, ages 0-18 months*<br>Conditional logistic regression | Matching on birth year and telephone area, adjustment for maternal age, education and income |
| Salonen, 1976 (20) | Glucocorticoids/*2*<br>*Routinely collected health data* | Classification: ICD<br>Main outcomes: Leukemias/*373*, brain tumors/*245*, other tumors/*354*<br>Secondary outcomes: Subgroups of the "other tumors"-category; kidney tumors/*96*, eye tumors/*37*, bone tumors/*56* (only reported for most commonly used medications) | Case-control/*1944*<br>*Finland, ages 0-14 years*<br>MacNemar's test, descriptive for rarely used medications | Matching on date of birth |
| Shaw, 2004 (23) | Immunosuppressants/*19*<br>*Maternal retrospective report* | Classification: ICD<br>ALL/*789* | Case-control/*1578*<br>*Canada, age 0-14 years*<br>Conditional logistic regression | Matching on sex and age at diagnosis, adjustment for maternal age and education |

| Reference<br>First author, year | Exposure<br>Medication/*exposed*<br>*Ascertainment*<br>Exposure window: during pregnancy, unless otherwise stated | Outcome<br>Classification system, type of cancer/*cases* | Materials and methods<br>Design/*sample size*<br>*Study characteristics*<br>Statistical analysis | <br>Method to account for confounding in the main analysis |
|---|---|---|---|---|
| Wen, 2002 (27) | Immunosuppressants or steroids in the year before pregnancy, through pregnancy and/or during breastfeeding/*28*<br>*Maternal retrospective report* | Classification system not stated<br>ALL/*1842* | Case-control/*3828*<br>*US, ages 0-14 years*<br>Conditional logistic regression | Matching on age, ethnicity, and telephone area, adjustment for parental income, education, ethnicity, smoking and alcohol use |
| LAXATIVES/ANTIDIARRHEALS | | | | |
| Birch, 1990 (92) | Laxatives/*not reported*<br>*Maternal retrospective report, validated in prescription records* | Classification: ICD<br>CNS tumors/*78* | Case-control/*234*<br>*UK, IRESCC, ages 0-15 years*<br>Mantel-Haenszel | Matching on age and sex |
| Gilman, 1989 (6) | Laxatives/*47*<br>*Maternal retrospective report and medical records. Children are considered exposed if medication use is present in either source* | Classification system not stated<br>Any cancer/*8059* | Case-control/*16,118*<br>*UK, OSCC, age not specified, only that cases are childhood deaths from cancer*<br>Conditional logistic regression | Matching on date of birth, sex and district, adjustment for abdominal x-rays in pregnancy, birth order among live born siblings, maternal age, and socioeconomic position |
| Hartley, 1988 (8) | Laxatives/*not reported*<br>*Maternal retrospective report, validated in prescription data* | Classification system not stated<br>Bone or soft tissue sarcomas/*73* | Case-control/*219*<br>*UK, IRESCC, ages 0-15 years*<br>Mantel-Haenszel, 95% CI by Cornfield's formula | Matching. Matching factors not reported, but they refer to a paper where methods are described (age and sex) |
| Ross, 1996 (57) | Laxatives/*13*<br>Antidiarrheals/*10*<br>*Maternal retrospective report* | Classification system not stated<br>IL/*84* | Case-control/*181*<br>*US, ages 0-1 year*<br>Conditional logistic regression | Matching on year of birth, geography and ethnicity, adjustment for maternal education |
| Ross, 2003 (19) | Loperamide/*7*<br>*Routinely collected health data* | Classification system not stated<br>IL/*243*, ALL/*157*, AML/*77* | Case-control/*636*<br>*US, ages 0-18 months*<br>Conditional logistic regression | Matching on birth year and telephone area, adjustment for maternal age, education and income |
| Wen, 2002 (27) | Laxatives or antidiarrheal agents in the year before pregnancy, through pregnancy and/or during breastfeeding/*75*<br>*Maternal retrospective report* | Classification system not stated<br>ALL/*1842* | Case-control/*3828*<br>*US, ages 0-14 years*<br>Conditional logistic regression | Matching on age, ethnicity, and telephone area, adjustment for parental |

| Reference | Exposure | Outcome | Materials and methods | |
|---|---|---|---|---|
| First author, year | Medication/*exposed*<br>*Ascertainment*<br>Exposure window: during pregnancy, unless otherwise stated | Classification system, type of cancer/*cases* | Design/*sample size*<br>*Study characteristics*<br>Statistical analysis | Method to account for confounding in the main analysis |
| | | | | income, education, ethnicity, smoking and alcohol use |
| ORAL CONTRACEPTIVES/SEX HORMONES | | | | |
| Ajrouche, 2014 (59) | Oral contraceptives/*116*, 1st generation/*1*, 2nd generation/*66*, 3rd generation/*12*, 4th generation/*1*, Other oral contraceptive/*10*<br>Also analysis by dose<br>*Maternal retrospective report, validated in a sensitivity analysis assuming worst case differential misclassification* | Classification system not stated<br>Main outcome: Acute leukemias/*747*<br>Secondary outcomes: ALL/*636*, AML/*100* | Case-control/*2168*<br>*France, ESCALE, ages 0-15 years*<br>Unconditional logistic regression | Matching on age and sex. Adjustment for age, sex, birth order, maternal age, education or occupation |
| Birch, 1990 (92) | Oral contraceptives any time before pregnancy/*not reported*<br>*Maternal retrospective report, validated in prescription records* | Classification: ICD<br>CNS tumors/*78* | Case-control/*234*<br>*UK, IRESCC, ages 0-15 years*<br>Mantel-Haenszel | Matching on age and sex |
| Bonaventure, 2015 (1) | Any sex hormones (ATC group G)/*426*<br>*Routinely collected health data* | Classification: ICD<br>ALL/*725*, AML/*128*, NHL/*83*, HL/*31*, astrocytoma/*100*, medulloblastoma/*48*, rhabdomyosarcoma/*54*, Ewings sarcoma/*20*, neuroblastoma/*78*, nephroblasoma/*60* | Case-control/*4122*<br>*UK, UKCCS, mean age 5.8 (max age 15 years)*<br>Unconditional logistic regression | Matching on sex, month and year of birth, and region of residence. Adjustment for year of birth |
| Buckley, 1989 (33) | Oral contraceptives in the year before pregnancy or during pregnancy/*39*<br>*Maternal retrospective report* | Classification system not stated<br>Hepatoblastoma/*75* | Case-control/*150*<br>*US and Canada, CCSG, upper age limit not stated, but the majority of children were under 2 years of age*<br>Conditional logistic regression | Matching on age |
| Bunin, 1987 (38) | Sex hormones in the year prior to birth/*15*<br>*Maternal retrospective report* | Classification system not stated<br>Nephroblastoma/*88*, also stratified by whether the case was genetic | Case-control/*176*<br>*US, ages 0-15 years*<br>Conditional logistic regression | Matching on telephone area code, year of birth and ethnicity. Adjustment for demographic characteristics (not specified further) |

| Reference | Exposure | Outcome | Materials and methods | |
|---|---|---|---|---|
| First author, year | Medication/*exposed*<br>*Ascertainment*<br>Exposure window: during pregnancy, unless otherwise stated | Classification system, type of cancer/*cases* | Design/*sample size*<br>*Study characteristics*<br>Statistical analysis | Method to account for confounding in the main analysis |
| Bunin, 1989 (34) | Oral contraceptives at conception/*5*<br>*Maternal retrospective report* | Classification system not stated<br>Nonheritable retinoblastoma/*115*, sporadic heritable retinoblastoma (bilateral cases or unilateral cases with 13q chromosomal deletion)/*67*<br>Cases with family history were excluded | Case-control/*383*<br>*US and Canada, CCSG, upper age limit not stated*<br>Conditional logistic regression | Matching on telephone area code, year of birth and ethnicity. Adjustment for paternal education |
| Bunin, 1994 (2) | Oral contraceptives /*20*<br>*Maternal retrospective report* | Classification system not stated<br>Astrocytoma/*155*, PNET/*166* | Case-control/*332*<br>*US and Canada, Children's Cancer Group, ages 0-5 years*<br>Conditional logistic regression | Matching on telephone area code, year of birth and ethnicity. Adjustment for income (only for astrocytoma) |
| Cardy, 2006 (35) | Sex hormones/*179*<br>*Maternal retrospective report* | Classification: ICD-O<br>Main outcome:<br>Any brain tumor/*1218*<br>Secondary outcomes:<br>Astroglial/*623*, PNET/*259*, other glial/*327* by morphological codes | Case-control/*3441*<br>*Europe, North America, Australia, SEARCH, upper age limit 19 years*<br>Unconditional logistic regression | Matching, not further specified. Adjustment for center, age, sex, year of birth, and parental education. Vitamin supplementation (unclear whether in mothers or children) was considered as both a confounder and effect modifier |
| Cook, 2004 (4) | Ethinyl estradiol/*27*, Mestranol/*17*, Norethindrone/*31*<br>Each of the above during pregnancy or breastfeeding<br>*Maternal retrospective report* | Classification system not stated<br>Neuroblastoma/*504* | Case-control/*1008*<br>*US and Canada, Children's Cancer Group, upper age limit 18 years*<br>Conditional logistic regression | Matching on age. Adjustment for maternal age, education, and ethnicity |
| Erjaee, 2017 (93) | Oral contraceptives in the first 2 months of pregnancy/*20*<br>*Maternal retrospective report, validated in medical records* | Classification system not stated<br>Any cancer/*300* | Case-control/*900*<br>*Iran, ages 0-17 years*<br>Logistic regression | Matching on age and sex. Adjustment for paternal education, family history of cancer, infection during pregnancy, hyperemesis gravidarum, parental |

| Reference<br>First author, year | Exposure<br>Medication/*exposed*<br>*Ascertainment*<br>Exposure window: during pregnancy, unless otherwise stated | Outcome<br>Classification system, type of cancer/*cases* | Materials and methods<br>Design/*sample size*<br>*Study characteristics*<br>Statistical analysis | <br>Method to account for confounding in the main analysis |
|---|---|---|---|---|
| | | | | smoking, radiation during pregnancy, child allergy, and environmental exposures for which it is unclear whether they take place before or after birth, including: use of wood burners at home, residence near power line, contact with insecticides, and contact with domestic animals |
| Gholami, 2011 (94) | Oral contraceptives any time before pregnancy/*108*<br>*Maternal retrospective report* | Classification system not stated<br>Acute leukemia/*130* | Case-control/*260*<br>*Iran, ages 0-15 years, study reported results for both population- and hospital controls, here we report the population controls*<br>Logistic regression | Matching on age and sex. Adjustment for maternal education and parental relationship status |
| Gilman, 1989 (6) | Sex hormones/*357*<br>*Maternal retrospective report and medical records. Children are considered exposed if medication use is present in either source* | Classification system not stated<br>Any cancer/*8059* | Case-control/*16,118*<br>*UK, OSCC, age not specified, only that cases are childhood deaths from cancer*<br>Conditional logistic regression | Matching on date of birth, sex and district. Adjustment for abdominal x-rays in pregnancy, birth order among live born siblings, maternal age, and socioeconomic position |
| Gold, 1979 (95) | Oral contraceptives any time before birth/*not reported*<br>Sex hormones/*not reported*<br>*Maternal retrospective report, uses cancer controls in addition to population controls, where the cancer controls are expected to be influenced by the same recall factors as cases* | Classification system not stated<br>Brain tumors/*73* | Case-control/*134*<br>*US, two control groups: healthy and cancer controls, ages 0-19 years*<br>Matched pairs OR, binomial test, exact 95% CI calculated based on the binomial distribution | Matching on ethnicity, sex, and year of birth |

| Reference | Exposure | Outcome | Materials and methods | |
|---|---|---|---|---|
| First author, year | Medication/*exposed*<br>*Ascertainment*<br>Exposure window: during pregnancy, unless otherwise stated | Classification system, type of cancer/*cases* | Design/*sample size*<br>*Study characteristics*<br>Statistical analysis | Method to account for confounding in the main analysis |
| Golding, 1990 (96) | Oral contraceptives in the 18 months before conception/*37*<br>*Maternal prospective report* | Classification system not stated<br>Any cancer/*33* | Case-cohort/*132*<br>*UK, ages 0-10 years*<br>Analysis after Pike and Morrow | Matching on maternal age, parity, marital status, paternal occupation, and plurality, each determined at delivery |
| Grufferman, 1982 (7) | Oral contraceptives or other birth control methods/*31,* Drugs for menstrual disorders or infertility/*7*<br>Each of the above 1 year prior to pregnancy or during pregnancy<br>*Maternal retrospective report* | Classification system not stated<br>Rhabdomyosarcoma/*33* | Case-control/*132*<br>*US, ages 0-14 years*<br>"Unmatched methods", not specified further | Matching on age, sex, and ethnicity |
| Hargreave, 2018 (97) | Any hormonal contraception/*914866*, Combined oral contraceptives/*781,833*, Combined non-oral contraceptives/*29,996*,<br>Progestin-only oral contraceptives/*77,152*,<br>Progestin-only non-oral contraception/*17,129*,<br>Emergency contraception/*131,503*<br>Each of the above >3 months prior to pregnancy, or ≤3 months prior or during pregnancy<br>*Routinely collected health data* | Classification: ICCC<br>Main outcome: Any leukemias/*606*<br>Secondary outcomes: Lymphoid leukemias/*465*, non-lymphoid leukemias/*141*, and genetic subtypes of leukemia | Cohort/*1,185,157*<br>*Denmark, ages 0-18 years*<br>Cox regression | Adjustment for year of birth, maternal age and infertility. |
| Hartley, 1988 (8) | Any sex hormones, including oral contraceptives/*not reported*<br>*Maternal retrospective report, validated in prescription data* | Classification system not stated<br>Bone or soft tissue sarcomas/*73* | Case-control/*219*<br>*UK, IRESCC, ages 0-15 years*<br>Mantel-Haenszel, 95% CI by Cornfield's formula | Matching. Matching factors not reported, but they refer to a paper where methods are described (age and sex) |
| Heck, 2015 (9) | Oral contraceptives/*131,* Non-oral hormonal contraceptives/*29*,<br>Any hormonal contraceptive/*154*<br>Each of the above in the year before pregnancy<br>*Maternal retrospective report* | Classification system not stated<br>Sporadic retinoblastoma, unilateral/*187*, sporadic retinoblastoma, bilateral/*95* | Case-control/*426*<br>*US/Canada, ages 0-15 years*<br>Unconditional logistic regression | Matching on year of birth. Adjustment for maternal ethnicity, education, income, age, and smoking in pregnancy |
| Herbst, 1971 (98) | Diethylstilbestrol/*7*<br>*Maternal retrospective report* | Classification system not stated<br>Vaginal adenocarcinoma/*8* | Case-control/*40*<br>*Boston, ages 14-22, two out of eight cases were >19 years*<br>Matched control method from Pike & Morrow | Matching on date of birth and type of delivery service (ward or private) |

| Reference | Exposure | Outcome | Materials and methods | |
|---|---|---|---|---|
| First author, year | Medication/*exposed*<br>*Ascertainment*<br>Exposure window: during pregnancy, unless otherwise stated | Classification system, type of cancer/*cases* | Design/*sample size*<br>*Study characteristics*<br>Statistical analysis | Method to account for confounding in the main analysis |
| Johnston, 1986 (99) | Oral contraceptives any time before pregnancy/*68*,<br>Oral contraceptives in the year prior to pregnancy/*54*<br>*Maternal retrospective report, validated in medical records* | Classification: According to a 1981 paper by Marsden et al.<br>*Germ cell tumors of childhood* (higher level of detail than the ICCC)<br>Germ cell tumors/*41* | Case-control/*123*<br>*UK, IRESCC, ages 0-15 years*<br>Descriptive statistics | Matching. Matching factors not reported, but they refer to a paper where methods are described (age and sex) |
| Kinnier-Wilson, 1981 (100) | Any hormones/*444*<br>*Maternal retrospective report, asks antenatal clinics to verify exposure* | Classification system not stated<br>Any cancers/*2287* | Case-control/*4317*<br>*UK, age not specified, only that cases are childhood deaths from cancer, results are stratified by x-ray exposure, we report results for x-ray unexposed*<br>Mantel-Haenszel | Matching, unclear on which factors, stratified on prenatal x-ray exposure |
| Kramer, 1987 (11) | Sex hormones 3 months prior to pregnancy/*not reported*<br>Sex hormones/*not reported*<br>*Maternal retrospective report, validated in medical records for a subsample. Validity compared between cases and controls* | Classification system not stated<br>Neuroblastoma/*104* | Case-control/*205*<br>*US, median age 1 year*<br>90% CI calculated by Miettinen's technique | Matching on telephone area code, ethnicity, and year of birth |
| Kwan, 2007 (44) | Oral contraceptives 3 months before and/or during pregnancy/*136*<br>*Maternal retrospective report* | Classification: ICD-O<br>Main outcome: Leukemia/*365*<br>Secondary outcome: ALL/*311* | Case-control/*825*<br>*US, NCCLS, ages 0-14 years*<br>Conditional logistic regression | Matching on date of birth, sex, ethnicity (of child and mother), and for part of the recruitment also county of residence at birth. Adjustment for maternal age, income, education |
| Linet, 1996 (101) | Oral contraceptives >3 months prior to pregnancy/*105*<br>Oral contraceptives ≤3 months prior to pregnancy/*98*<br>*Routinely collected health data* | Classification: The American Cancer Society Revision of the World Health Organization Classification of Brain Tumors for childhood brain tumors | Case-control/*624*<br>*Sweden, ages 0-17 years*<br>Conditional logistic regression | Matching on sex, birth year and month. Adjustment for matching factors, gestational age and maternal narcotic use |

| Reference | Exposure | Outcome | Materials and methods | |
|---|---|---|---|---|
| First author, year | Medication/*exposed*<br>*Ascertainment*<br>Exposure window: during pregnancy, unless otherwise stated | Classification system, type of cancer/*cases* | Design/*sample size*<br>*Study characteristics*<br>Statistical analysis | Method to account for confounding in the main analysis |
| | | Main outcome: Any brain tumor/*104*<br>Secondary outcomes: Low grade astrocytoma/*37*, high grade astrocytoma/*7*, medulloblastoma/*17*, *e*pendymoma/*15*, others/*27* | | |
| Lupo, 2014 (102) | Any birth control (unspecified) any time before pregnancy/*463*<br>Any birth control (unspecified) at conception/*22*<br>*Maternal retrospective report* | Classification system not stated<br>Main outcome: Rhabdomyosarcoma/*322*<br>Secondary outcomes: Embryonal rhabdomyosarcoma/*215*, alveolar rhabdomyosarcoma/*66* | Case-control/*644*<br>*US, Children's Oncology Group, ages 0-20 years*<br>Conditional logistic regression | Matching on ethnicity, sex, and age. Adjustment for matching factors, parental education and income |
| McCredie, 1994 (13) | Oral contraceptives/*25*<br>Drugs to prevent miscarriage/*8*<br>Each of the above in the month before or during pregnancy<br>*Maternal retrospective report* | Classification: ICD<br>Brain or cranial nerves tumors/*82* | Case-control/*246*<br>*Australia, ages 0-14 years*<br>Conditional logistic regression | Matching on sex and age. Adjustment for parental education |
| McKinney, 1987 (14) | Hormones/*not reported*<br>Hormonal confirmation of pregnancy/*not reported*<br>*Maternal retrospective report and medical records, findings from the two data sources are reported separately* | Classification system not stated<br>Main outcome: Leukemias/lymphomas/*234*<br>Secondary outcomes: Leukemias/*171*<br>Only reported separately for significant associations | Case-control/*702*<br>*UK, IRESCC, ages 0-15 years*<br>Mantel-Haenszel | Matching on age and sex |
| Michalek, 1996 (29) | Sex hormones, any/*24*<br>Sex hormones by indication: Infertility/*6*, Vaginal bleeding/*8*, To maintain pregnancy/*13*, Previous miscarriage/*3*<br>*Maternal retrospective report* | Classification system not stated<br>Neuroblastoma/*183* | Case-control/*555*<br>*US, ages 0-14 years*<br>Unconditional logistic regression | Matching on year of birth |
| Olshan, 1999 (73) | Oral contraceptives/*278* | Classification system not stated | Case-control/*1008* | Matching on date of birth. Adjustment for maternal |

| Reference<br>First author, year | Exposure<br>Medication/*exposed*<br>*Ascertainment*<br>Exposure window: during pregnancy, unless otherwise stated | Outcome<br>Classification system, type of cancer/*cases* | Materials and methods<br>Design/*sample size*<br>*Study characteristics*<br>Statistical analysis | <br>Method to account for confounding in the main analysis |
|---|---|---|---|---|
| | Ever, in the 2-12 months prior to pregnancy, in the month prior to pregnancy, or during the first trimester<br>*Maternal retrospective report* | Neuroblastoma/*504*, also neuroblastoma subgroups by MYCN oncogene amplification status in an exploratory analysis | *US and Canada, ages 0-19 years*<br>Conditional logistic regression | ethnicity, education, and income |
| Operskalski, 1987 (103) | Estrogen in the year before or during pregnancy/*not reported*<br>*Maternal retrospective report* | Classification system not stated<br>Osteosarcoma/*64* | Case-control/*188*<br>*US, ages 0-25 years (88% of the population were under 20 years old)*<br>Breslow & Day's method | Matching on ethnicity, sex, and birth year |
| Ou Shu, 2002 (104) | Oral contraceptive/*126*, Hormone use other than oral contraceptives (unspecified) in the year prior to pregnancy/*173*, Hormone use other than oral contraceptives (unspecified)/*87*<br>*Maternal retrospective report* | Classification system not stated<br>Main outcome: ALL/*1842*<br>Secondary outcomes: T-cell ALL/*183*, early pre-B-cell ALL/*893*, pre-B-cell ALL/*233* | Case-control/*3828*<br>*US, ages 0-14 years*<br>Conditional logistic regression | Matching on age, ethnicity, telephone area code, and for T-cell ALL also sex. Adjustment for maternal education, ethnicity, age, and income |
| Pombo de Oliveira, 2006 (17) | Abortive drugs/*130*, Misoprostol/*13*, Hormones (Oral contraceptives, antiabortive progesterone treatment, and thyroid hormones)/*22*<br>Each of the above 3 months before pregnancy, during pregnancy, and/or during breastfeeding<br>*Maternal retrospective report* | Classification system not stated<br>Infant acute leukemias/*202*, also analyzed by MLL gene translocation status | Case-control/*642*<br>*Brazil, ages 0-21 months*<br>Unconditional logistic regression | Matching on age and region. Adjustment for region, sex, birth weight, maternal age, and income |
| Puumala, 2007 (75) | Ever use of oral contraceptives/*252*, Ever use of non-oral hormonal contraceptives/*20*<br>*Maternal retrospective report* | Classification system not stated<br>Main outcome: Leukemias/*158*<br>Secondary outcomes: ALL/*97*, AML/*61* | Case-control/*331*<br>*US, Children's Oncology Group, ages 0-19 years, all cases and controls have Down syndrome.*<br>Unconditional logistic regression | Matching on age. Adjustment for maternal age, ethnicity, education, and child sex |
| Robison, 1989 (37) | Oral contraceptives/*130*<br>Hormones (hormones of any sort, such as female hormones, fertility drugs, cortisone, insulin or thyroid hormones)/*23* | Classification system not stated<br>ANLL/*204* | Case-control/*408*<br>*US, ages 0-18 years*<br>Conditional logistic regression | Matching on date of birth, ethnicity, and telephone area code, stated that they adjust for confounders, but these are not specified |

| Reference<br>First author, year | Exposure<br>Medication/*exposed*<br>*Ascertainment*<br>Exposure window: during pregnancy, unless otherwise stated | Outcome<br>Classification system, type of cancer/*cases* | Materials and methods<br>Design/*sample size*<br>*Study characteristics*<br>Statistical analysis | <br>Method to account for confounding in the main analysis |
|---|---|---|---|---|
| | Each of the above in the year before pregnancy, through pregnancy and/or during breastfeeding<br>Also analyses by duration and frequency<br>*Maternal retrospective report, validated by comparisons to associations in four other independent case-control studies where cases of different cancers did not report more medication use than controls* | | | |
| Salonen, 1976 (20) | Estrogen/*9*<br>Progesterone/*37*<br>*Routinely collected health data* | Classification: ICD<br>Main outcomes: Leukemias/*373*, brain tumors/*245*, other tumors/*354*<br>Secondary outcomes: Subgroups of the "other tumors"-category; kidney tumors/*96*, eye tumors/*37*, bone tumors/*56* (only reported for most commonly used medications) | Case-control/*1944*<br>*Finland, ages 0-14 years*<br>MacNemar's test, descriptive for rarely used medications | Matching on date of birth |
| Schüz, 2001 (82) | Oral contraceptives or sex hormones/*38*<br>*Maternal retrospective report from questionnaire data validated through phone interview* | Classification: International Neuroblastoma Staging System<br>Neuroblastoma/*163* | Case-control/*1860*<br>*Germany, ages 0-7 years*<br>Frequency matched conditional logistic regression | Matching on sex, date of birth, and district. Adjustment for age, sex, year of birth, degree of urbanization, and socioeconomic position |
| Shankar, 2006 (105) | Female hormones six months before pregnancy until delivery, according to self-report/*53*, according to medical records/*123*<br>*Maternal retrospective report and medical records, findings from the two data sources are reported separately* | Classification system not stated<br>Germ cell tumors of all other sites than the brain/*278* | Case-control/*701*<br>*US, Children's Oncology Group, ages 0 to 14 years*<br>Unconditional logistic regression | Matching on age, sex, and state. Adjustment for matching factors, maternal ethnicity, education and income |
| Sharpe & Franco, 1996 (22) | Oral contraceptives/*30*<br>*Maternal retrospective report* | Classification system not stated<br>Nephroblastoma/*109* | Case-control/*327*<br>*Brazil, age range not reported, mean age 3 years and 5 months* | Matching on age, sex, and interviewer, stratified on socioeconomic position |

| Reference | Exposure | Outcome | Materials and methods | |
|---|---|---|---|---|
| First author, year | Medication/*exposed*<br>*Ascertainment*<br>Exposure window: during pregnancy, unless otherwise stated | Classification system, type of cancer/*cases* | Design/*sample size*<br>*Study characteristics*<br>Statistical analysis | Method to account for confounding in the main analysis |
| | | | Logistic regression | |
| Shaw, 2004 (23) | Oral contraceptive/*19*<br>*Maternal retrospective report* | Classification: ICD<br>ALL/*789* | Case-control/*1578*<br>*Canada, age 0-14 years*<br>Conditional logistic regression | Matching on sex and age at diagnosis. Adjustment for maternal age and education |
| Shu, 1995 (24) | Birth control pills/*27*<br>*Maternal retrospective report* | Classification system not stated<br>Malignant germ-cell tumors/*105* | Case-control/*744*<br>*US and Canada, CCG (Children's Cancer Group), ages 0-15 years*<br>Unconditional logistic regression | Matching on telephone area code. Adjustment for child age, sex, gestational age, number of siblings, maternal education, and smoking |
| Stewart, 1958 (106) | Chemical contraceptives before birth of the child (ever before or during pregnancy)/*27*<br>*Maternal retrospective report, validated through a comparison of data completeness for sensitive questions between cases and controls* | Classification system not stated<br>Leukemia/*677*, other cancers/*739* | Case-control/*2301*<br>*England and Wales, ages 0-9 years*<br>Comparison of matched children, not specified further | Matched on age, sex and locality, or chosen at random. Not clear how many were chosen in each fashion |
| Stålberg, 2010 (25) | Hormones (unspecified)/*<10*<br>*Routinely collected health data* | Classification: ICD<br>Brain tumor (ICD-7 code 193)/*512* | Case-control/*1037*<br>*Sweden, ages 0-15 years*<br>Logistic regression | Matching on sex and birth year. Adjustment for maternal age, parity, country of birth, and level of hospital (primary, secondary, or tertiary) |
| Tournaire, 2016 (107) | Maternal in-utero exposure to diethylstilbestrol/*4409*<br>*Maternal retrospective report, validated by comparison of prevalence to the prevalence in the general population in Europe* | Classification: ICD<br>Any cancer/*25* | Cohort/*10,612*<br>*France, ages 0-43 years (women average: 15.2, men average: 15.5)*<br>Logistic regression | Adjustment for year of birth |
| Van Duijn, 1994 (52) | Drugs to maintain pregnancy in the year before or during pregnancy/*7*<br>*Maternal retrospective report, validated by comparison to associations in another case-control study where cases of a different type of cancer did not report more medication use than controls* | Classification system not stated<br>ANLL/*80* | Case-control/*320*<br>*The Netherlands, ages 0-14 years, study reported results for both population- and cancer controls*<br>Conditional logistic regression | Matching on sex and age. Adjustment for sex, year of birth, maternal socioeconomic position, and in pregnancy: alcohol use, smoking, occupational hydrocarbon exposure, other medication |

| Reference<br>First author, year | Exposure<br>Medication/*exposed*<br>*Ascertainment*<br>Exposure window: during pregnancy, unless otherwise stated | Outcome<br>Classification system, type of cancer/*cases* | Materials and methods<br>Design/*sample size*<br>*Study characteristics*<br>Statistical analysis | Method to account for confounding in the main analysis |
|---|---|---|---|---|
| | | | | use, ultrasound, x-rays, viral infections |
| Van Steensel-Moll, 1985 (53) | Drugs to maintain pregnancy/*44*<br>*Maternal retrospective report* | Classification: ICD<br>Leukemias/*519* | Case-control/*1026*<br>*The Netherlands, ages 0-15 years*<br>Logistic regression | Matching on date of birth, sex and municipality. Adjustment for age and sex |
| Wen, 2002 (27) | Oral contraceptives/*123*<br>Hormones (not specified)/*54*<br>Each in the year before pregnancy, through pregnancy and/or during breastfeeding<br>*Maternal retrospective report* | Classification system not stated<br>ALL/*1842* | Case-control/*3828*<br>*US, ages 0-14 years*<br>Conditional logistic regression | Matching on age, ethnicity, and telephone area. Adjustment for parental income, education, ethnicity, smoking and alcohol use |
| PSYCHOTROPHICS | | | | |
| Birch, 1990 (92) | Antiepileptics/*not reported*<br>Sedatives or anxiolytics/*not reported*<br>*Maternal retrospective report, validated in prescription records* | Classification: ICD<br>CNS tumors/*78* | Case-control/*234*<br>*UK, IRESCC, ages 0-15 years*<br>Mantel-Haenszel | Matching on age and sex |
| Buckley, 1989 (33) | Antiepileptics/*0*, Sedatives or anxiolytics/*5*<br>Each of the above in the year before pregnancy or during pregnancy<br>*Maternal retrospective report* | Classification system not stated<br>Hepatoblastoma/*75* | Case-control/*150*<br>*US and Canada, CCSG, upper age limit not stated, but the majority of children were under 2 years of age*<br>Conditional logistic regression | Matching on age |
| Cardy, 2006 (35) | Antiepileptics/*50*, Barbiturates/*30*<br>*Maternal retrospective report* | Classification: ICD-O<br>Main outcome:<br>Any brain tumor/*1218*<br>Secondary outcomes:<br>Astroglial/*623*, PNET/*259*, other glial/*327* by morphological codes | Case-control/*3441*<br>*Europe, North America, Australia, SEARCH, upper age limit 19 years*<br>Unconditional logistic regression | Matching, not further specified. Adjustment for center, age, sex, year of birth, and parental education. Vitamin supplementation (unclear whether in mothers or children) was considered as both a confounder and effect modifier |

| Reference | Exposure | Outcome | Materials and methods | |
|---|---|---|---|---|
| First author, year | Medication/*exposed*<br>*Ascertainment*<br>Exposure window: during pregnancy, unless otherwise stated | Classification system, type of cancer/*cases* | Design/*sample size*<br>*Study characteristics*<br>Statistical analysis | Method to account for confounding in the main analysis |
| Carozza, 1995 (3) | Amitriptyline/*1*, Amitriptyline HCL/*2*, Chlordiazepoxide HCL/*8*<br>Imipramine HCL/*1*, Nortriptyline HCL/*1*, Thiothixene/*1*<br>*Maternal retrospective report* | Classification: Histological Typing of Tumors of the Central Nervous System. International Histological Classification of Tumors<br>Any brain tumors/*361* | Case-control/*1444*<br>*US, SEER, upper age limit 18 years*<br>Conditional logistic regression, for main exposure (any n-nitrosatable drugs), descriptive statistics for medications stated here | Matching on age, sex and maternal ethnicity |
| Cordier, 1994 (36) | Barbiturates/*2*<br>*Maternal retrospective report* | Classification system not stated<br>Brain tumors/*75* | Case-control/*188*<br>*France, ages 0-15 years*<br>Conditional logistic regression | Matching on birth year. Adjustment for age, sex and maternal age |
| Gilman, 1989 (6) | Antidepressants/*93*, Antiepileptics/*37*, Hypnotics/*323*, Sedatives or anxiolytics/*1269*<br>*Maternal retrospective report and medical records. Children are considered exposed if medication use is present in either source* | Classification system not stated<br>Any cancer/*8059* | Case-control/*16,118*<br>*UK, OSCC, age not specified, only that cases are childhood deaths from cancer*<br>Conditional logistic regression | Matching on date of birth, sex and district. Adjustment for abdominal x-rays in pregnancy, birth order among live born siblings, maternal age, and socioeconomic position |
| Gold, 1978 (108) | Barbiturates/*6*<br>*Maternal retrospective report, uses cancer controls in addition to population controls, where the cancer controls are expected to be influenced by the same recall factors as cases* | Classification system not stated<br>Brain tumors/*73* | Case-control/*136*<br>*US, two control groups: healthy and cancer controls, we report healthy controls, ages 0-19 years*<br>Matched pairs OR, binomial test, exact 95% CI calculated based on the binomial distribution | Matching on ethnicity, sex, date and age of diagnosis |
| Goldhaber, 1990 (109) | Barbiturates/*58*<br>*Routinely collected health data* | Classification system not stated<br>Intracranial and spinal cord tumors/*86* | Case/control/*258*<br>*US, ages 0-19 years*<br>Conditional logistic regression | Matching on year of birth, sex, date of entry into the medical care program, birth at medical care program hospital |
| Grufferman, 1982 (7) | Anxiolytics 1 year prior to or during pregnancy/*7*<br>*Maternal retrospective report* | Classification system not stated<br>Rhabdomyosarcoma/*33* | Case-control/*132*<br>*US, ages 0-14 years* | Matching on age, sex, and ethnicity |

| Reference<br>First author, year | Exposure<br>Medication/*exposed*<br>*Ascertainment*<br>Exposure window: during pregnancy, unless otherwise stated | Outcome<br>Classification system, type of cancer/*cases* | Materials and methods<br>Design/*sample size*<br>*Study characteristics*<br>Statistical analysis | <br>Method to account for confounding in the main analysis |
|---|---|---|---|---|
| | | | "Unmatched methods", not specified further | |
| Gurney, 1997 (110) | Antiepileptics 1 month before pregnancy, during pregnancy or during breastfeeding/*27*<br>*Maternal retrospective report* | Classification: ICD-O<br>Any brain tumors/*540* | Case-control/*1341*<br>*US, ages 0-19 years*<br>Unconditional logistic regression | Matching on geographic area, age, sex, and study site |
| Hartley, 1988 (8) | Sedatives or anxiolytics/*not reported*<br>*Maternal retrospective report, validated in prescription data* | Classification system not stated<br>Bone or soft tissue sarcomas/*73* | Case-control/*219*<br>*UK, IRESCC, ages 0-15 years*<br>Mantel-Haenszel, 95% CI by Cornfield's formula | Matching. Matching factors not reported, but they refer to a paper where methods are described (age and sex) |
| Howe, 1989 (111) | Barbiturates or phenytoin/*5*<br>*Maternal retrospective report* | Classification system not stated<br>Brain tumors/*74* | Case-control/*212*<br>*Canada, ages 0-19 years*<br>Conditional logistic regression | Matching on sex, year of birth, and area of residence |
| Kinnier-Wilson, 1981 (100) | Sedatives/*849*<br>*Maternal retrospective report, asks antenatal clinics to verify exposure* | Classification system not stated<br>Any cancers/*2287* | Case-control/*4317*<br>*UK, age not specified, only that cases are childhood deaths from cancer, results are stratified by x-ray exposure*<br>Mantel-Haenszel | Matching, unclear on which factors, stratified on prenatal x-ray exposure |
| Kramer, 1987 (11) | Anxiolytics/*not reported*<br>*Maternal retrospective report, validated in medical records for a subsample. Validity compared between cases and controls* | Classification system not stated<br>Neuroblastoma/*104* | Case-control/*205*<br>*US, median age 1 year*<br>90% CI calculated by Miettinen's technique | Matching on telephone area code, ethnicity, and year of birth |
| Kuijten, 1990 (12) | Neurally active medications/*not reported*<br>*Maternal retrospective report* | Classification system not stated<br>Astrocytoma/*163* | Case-control/*326*<br>*US, ages 0-14 years*<br>Conditional logistic regression | Matching on telephone area code, ethnicity, and age |
| McCredie, 1994 (13) | Hypnotics, sedatives or anxiolytics in the month before or during pregnancy/*12*<br>*Maternal retrospective report* | Classification: ICD<br>Brain or cranial nerves tumors/*82* | Case-control/*246*<br>*Australia, ages 0-14 years*<br>Conditional logistic regression | Matching on sex and age. Adjustment for parental education |
| McKinney, 1999 (15) | Hypnotics or anxiolytics/*34*<br>*Routinely collected health data* | Classification: ICCC<br>Main outcome: Leukemia/*144* | Case-control/*415*<br>*Scotland, ages 0-14 years*<br>Conditional logistic regression | Matching on age, area of residence, and sex |

| Reference<br>First author, year | Exposure<br>Medication/*exposed*<br>*Ascertainment*<br>Exposure window: during pregnancy, unless otherwise stated | Outcome<br>Classification system, type of cancer/*cases* | Materials and methods<br>Design/*sample size*<br>*Study characteristics*<br>Statistical analysis | <br>Method to account for confounding in the main analysis |
|---|---|---|---|---|
| | | Secondary outcome: ALL/*124*, lymphomas/*45*, CNS tumors/*75*, other solid tumors/*126* | | |
| Momen, 2018 (112) | Antidepressants from 2 years prior until 1 month prior to pregnancy/*30,607*, Antidepressants from 1 month prior to pregnancy until end of pregnancy/*21,488*<br>Comparators are women who used antidepressants prior to, but not during pregnancy. Also analysis by duration<br>*Routinely collected health data, validated in a sensitivity analysis restricted to women with two or more prescription fills* | Classification: ICD<br>Main outcome: Any cancer/*1298*<br>Secondary outcomes: Leukemias/*524*, CNS/SNS tumors/*236* | Cohort/*915,128*<br>*Denmark, ages 0-14 years*<br>Cox regression | Adjustment for maternal age, parity, history of psychiatric disorders, inpatient psychiatric treatment in the two years before pregnancy, psychotropic comedication during pregnancy, education, calendar year at birth of child, and parental history of cancer |
| Ross, 2003 (19) | Hydroxyzine/*13*<br>*Routinely collected health data* | Classification system not stated<br>IL/*243*, ALL/*157*, AML/*77* | Case-control/*636*<br>*US, ages 0-18 months*<br>Conditional logistic regression | Matching on birth year and telephone area. Adjustment for maternal age, education and income |
| Robison, 1989 (37) | Sedatives in the year before pregnancy, through pregnancy and/or during breastfeeding/*17*<br>Also analyses by duration and frequency<br>*Maternal retrospective report, validated by comparisons to associations in four other independent case-control studies where cases of different cancers did not report more medication use than controls* | Classification system not stated<br>ANLL/*204* | Case-control/*408*<br>*US, ages 0-18 years*<br>Conditional logistic regression | Matching on date of birth, ethnicity, and telephone area code, stated that they adjust for confounders, but these are not specified |
| Rodvall, 1990 (47) | Sedatives or sleeping pills (unspecified)/*28*<br>*Routinely collected health data* | Classification system not stated<br>Any cancer/*95* | Case-control/*285*<br>*Sweden, ages 0-16 years*<br>Multiple conditional logistic regression | Matching on sex and birth year and month. Adjustment for maternal age, drug use, obstetric complications, previous miscarriage, socioeconomic position and child gestational age. |

| Reference<br>First author, year | Exposure<br>Medication/*exposed*<br>*Ascertainment*<br>Exposure window: during pregnancy, unless otherwise stated | Outcome<br>Classification system, type of cancer/*cases* | Materials and methods<br>Design/*sample size*<br>*Study characteristics*<br>Statistical analysis | <br>Method to account for confounding in the main analysis |
|---|---|---|---|---|
| Roman, 1997 (48) | Anticonvulsants/*8*<br>*Routinely collected health data* | Classification system not stated<br>Main outcomes: Leukemias/*143*, NHL/*34*<br>Secondary outcomes: ALL/*113*, AML/*15* | Case-control/*429*<br>*England, ages 3 months to 29 years*<br>Conditional logistic regression | Matching on hospital catchment area of birth, sex, year and month of birth |
| Salonen, 1976 (20) | Neuroleptics/*7*, Antineurotics/*11*, Barbiturates/*29*<br>*Routinely collected health data* | Classification: ICD<br>Main outcomes: Leukemias/*373*, brain tumors/*245*, other tumors/*354*<br>Secondary outcomes: Subgroups of the "other tumors"-category; kidney tumors/*96*, eye tumors/*37*, bone tumors/*56* (only reported for most commonly used medications) | Case-control/*1944*<br>*Finland, ages 0-14 years*<br>MacNemar's test, descriptive for rarely used medications | Matching on date of birth |
| Sanders and Draper, 1979 (49) | Phenytoin/*18*, Phenobarbitone/*32*<br>*Maternal retrospective report and medical records, findings from the two data sources are reported separately* | Classification system not stated<br>Any cancer/*39* | Case-controls/*61*<br>*UK, Oxford Survey of Childhood Cancers, ages 0-15. All children were born to mothers with epilepsy*<br>Comparison of the frequencies | It would appear that matching was done, but it is unclear |
| Sharpe & Franco, 1996 (22) | Anticonvulsants/*5*<br>*Maternal retrospective report* | Classification system not stated<br>Nephroblastoma/*109* | Case-control/*327*<br>*Brazil, age range not reported, mean age 3 years and 5 months*<br>Logistic regression | Matching on age, sex, and interviewer, stratified on socioeconomic position |
| Shaw, 2004 (23) | CNS depressants (tranquilizers and sleeping pills)/*14*<br>Epileptic medication/*7*<br>*Maternal retrospective report* | Classification: ICD<br>ALL/*789* | Case-control/*1578*<br>*Canada, age 0-14 years*<br>Conditional logistic regression | Matching on sex and age at diagnosis. Adjustment for maternal age and education |
| Stålberg, 2010 (25) | Neuroleptics/*7*, Antiepileptics/*<10*, Antipsychotics/*<10*, Anxiolytics/*<10* | Classification: ICD<br>Brain tumor (ICD-7 code 193)/*512* | Case-control/*1037*<br>*Sweden, ages 0-15 years*<br>Logistic regression | Matching on sex and birth year. Adjustment for maternal age, parity, country of birth, |

| Reference<br>First author, year | Exposure<br>Medication/*exposed*<br>*Ascertainment*<br>Exposure window: during pregnancy, unless otherwise stated | Outcome<br>Classification system, type of cancer/*cases* | Materials and methods<br>Design/*sample size*<br>*Study characteristics*<br>Statistical analysis | <br>Method to account for confounding in the main analysis |
|---|---|---|---|---|
| | *Routinely collected health data* | | | and level of hospital (primary, secondary, or tertiary) |
| Van Steensel-Moll, 1985 (53) | Sedatives or sleeping pills/*21*<br>*Maternal retrospective report* | Classification: ICD<br>Leukemias/*519* | Case-control/*1026*<br>*The Netherlands, ages 0-15 years*<br>Logistic regression | Matching on date of birth, sex and municipality. Adjustment for age and sex |
| Wen, 2002 (27) | Sedatives, tranquilizers, or sleeping pills/*17*, Dilantin or other antiepileptic (seizure) drugs/*7*<br>Each of the above in the year before pregnancy, through pregnancy and/or during breastfeeding<br>*Maternal retrospective report* | Classification system not stated<br>ALL/*1842* | Case-control/*3828*<br>*US, ages 0-14 years*<br>Conditional logistic regression | Matching on age, ethnicity, and telephone area. Adjustment for parental income, education, ethnicity, smoking and alcohol use |
| SKIN TREATMENT | | | | |
| Bonaventure, 2015 (1) | Dermatologicals (ATC group D)/*472*<br>*Routinely collected health data* | Classification: ICD<br>ALL/*725*, AML/*128*, NHL/*83*, HL/*31*, astrocytoma/*100*, medulloblastoma/*48*, rhabdomyosarcoma/*54*, Ewings sarcoma/*20*, neuroblastoma/*78*, nephroblasoma/*60* | Case-control/*4122*<br>*UK, UKCCS, mean age 5.8 (max age 15 years)*<br>Unconditional logistic regression | Matching on sex, month and year of birth, and region of residence. Adjustment for year of birth |
| Gilman, 1989 (6) | Local skin treatment/*22*<br>*Maternal retrospective report and medical records. Children are considered exposed if medication use is present in either source* | Classification system not stated<br>Any cancer/*8059* | Case-control/*16,118*<br>*UK, OSCC, age not specified, only that cases are childhood deaths from cancer*<br>Conditional logistic regression | Matching on date of birth, sex and district. Adjustment for abdominal x-rays in pregnancy, birth order among live born siblings, maternal age, and socioeconomic position |
| Ross, 1996 (57) | Acne medications/*7*<br>*Maternal retrospective report* | Classification system not stated<br>IL/*84* | Case-control/*181*<br>*US, ages 0-1 year*<br>Conditional logistic regression | Matching on year of birth, geography and ethnicity. Adjustment for maternal education |
| THYROID THERAPY | | | | |

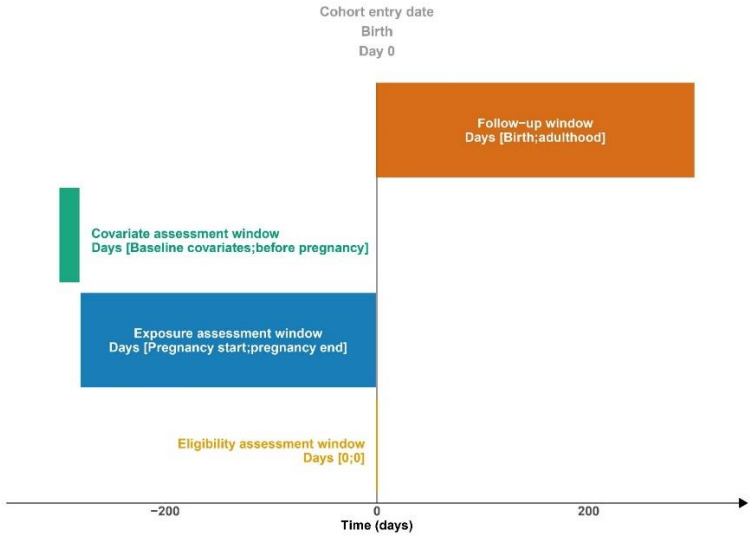
| Reference | Exposure | Outcome | Materials and methods | |
|---|---|---|---|---|
| First author, year | Medication/*exposed*<br>*Ascertainment*<br>Exposure window: during pregnancy, unless otherwise stated | Classification system, type of cancer/*cases* | Design/*sample size*<br>*Study characteristics*<br>Statistical analysis | Method to account for confounding in the main analysis |
| Cook, 2004 (4) | Levothyroxine during pregnancy or breastfeeding/*17*<br>*Maternal retrospective report* | Classification system not stated<br>Neuroblastoma/*504* | Case-control/*1008*<br>*US and Canada, Children's Cancer Group, upper age limit 18 years*<br>Conditional logistic regression | Matching on age. Adjustment for maternal age, education, and ethnicity |
| Gilman, 1989 (6) | Thyroid hormone/*20*<br>*Maternal retrospective report and medical records. Children are considered exposed if medication use is present in either source* | Classification system not stated<br>Any cancer/*8059* | Case-control/*16,118*<br>*UK, OSCC, age not specified, only that cases are childhood deaths from cancer*<br>Conditional logistic regression | Matching on date of birth, sex and district. Adjustment for abdominal x-rays in pregnancy, birth order among live born siblings, maternal age, and socioeconomic position |
| Ognjanovic, 2009 (30) | Thyroid medication/*15*<br>*Maternal retrospective report* | Classification system not stated<br>Main outcome: Leukemias/*158*<br>Secondary outcomes: ALL/*97*, AML/*61* | Case-control/*331*<br>*US, Children's Oncology Group, ages 0-19 years, all cases and controls have Down syndrome*<br>Descriptive statistics | Matching on age and sex |
| Ross, 2003 (19) | Levothyroxine/*8*<br>*Routinely collected health data* | Classification system not stated<br>IL/*243*, ALL/*157*, AML/*77* | Case-control/*636*<br>*US, ages 0-18 months*<br>Conditional logistic regression | Matching on birth year and telephone area. Adjustment for maternal age, education and income |
| Shaw, 2004 (23) | Hormones (fertility and thyroid hormones)/*31*<br>*Maternal retrospective report* | Classification: ICD<br>ALL/*789* | Case-control/*1578*<br>*Canada, age 0-14 years*<br>Conditional logistic regression | Matching on sex and age at diagnosis. Adjustment for maternal age and education |
| MISCELLANEOUS MEDICATIONS | | | | |
| Bonaventure, 2015 (1) | Blood and blood forming organs (ATC group B)/*1155*<br>*Routinely collected health data* | Classification: ICD<br>ALL/*725*, AML/*128*, NHL/*83*, HL/*31*, astrocytoma/*100*, medulloblastoma/*48*, rhabdomyosarcoma/*54*, Ewings | Case-control/*4122*<br>*UK, UKCCS, mean age 5.8 (max age 15 years)*<br>Unconditional logistic regression | Matching on sex, month and year of birth, and region of residence. Adjustment for year of birth |

| Reference | Exposure | Outcome | Materials and methods | |
|---|---|---|---|---|
| First author, year | Medication/*exposed*<br>*Ascertainment*<br>Exposure window: during pregnancy, unless otherwise stated | Classification system, type of cancer/*cases* | Design/*sample size*<br>*Study characteristics*<br>Statistical analysis | Method to account for confounding in the main analysis |
| | | sarcoma/*20*, neuroblastoma/*78*, nephroblasoma/*60* | | |
| Gilman, 1989 (6) | Gastro-intestinal sedatives/*43*, Anticoagulants/*6*, Vasodilators/*8*,<br>Other cardiac/*29*, Pituitary hormone/*186*, Local ear, nose, throat treatment/*39*, Local eye treatment/*9*<br>*Maternal retrospective report and medical records. Children are considered exposed if medication use is present in either source* | Classification system not stated<br>Any cancer/*8059* | Case-control/*16,118*<br>*UK, OSCC, age not specified, only that cases are childhood deaths from cancer*<br>Conditional logistic regression | Matching on date of birth, sex and district. Adjustment for abdominal x-rays in pregnancy, birth order among live born siblings, maternal age, and socioeconomic position |
| McKinney, 1987 (14) | Topical steroids/*not reported*<br>*Maternal retrospective report and medical records, findings from the two data sources are reported separately* | Classification system not stated<br>Main outcome: Leukemias/lymphomas/*234*<br>Secondary outcomes: Leukemias/*171*<br>Only reported separately for significant associations | Case-control/*702*<br>*UK, IRESCC, ages 0-15 years*<br>Mantel-Haenszel | Matching on age and sex |
| Michalek, 1996 (29) | Antipyretics/*20*<br>*Maternal retrospective report* | Classification system not stated<br>Neuroblastoma/*183* | Case-control/*555*<br>*US, ages 0-14 years*<br>Unconditional logistic regression | Matching on year of birth |
| Salonen, 1976 (20) | Papaverine/*13*<br>*Routinely collected health data* | Classification: ICD<br>Main outcomes: Leukemias/*373*, brain tumors/*245*, other tumors/*354*<br>Secondary outcomes: Subgroups of the "other tumors"-category; kidney tumors/*96*, eye tumors/*37*, bone tumors/*56* (only reported for most commonly used medications) | Case-control/*1944*<br>*Finland, ages 0-14 years*<br>MacNemar's test, descriptive for rarely used medications | Matching on date of birth |

[a]Studies on IVF or unspecified assisted reproduction have been included, as assisted reproduction will almost always include use of medications.
ALL: Acute lymphocytic leukemias, AML: Acute myeloid leukemias, ANLL: Acute non-lymphocytic leukemias, ATC: Anatomical Therapeutic Chemical Classification System, CI: Confidence Interval, CNS: Central Nervous System, GnRH: Gonadotropin-releasing hormone, hCG: human chorionic gonadotropin, HIV: Human immunodeficiency virus, HL:

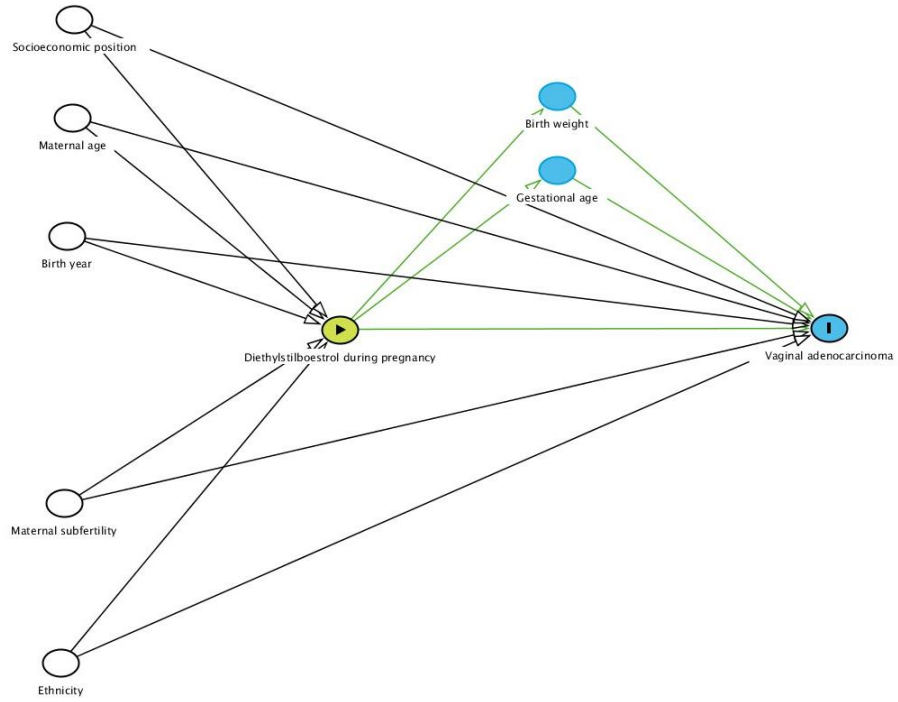
Hodgkin’s lymphoma, hMG: human menopausal gonadotropin, ICCC: International Classification of Childhood Cancers, ICD: International Classification of Diseases, ICD-O: International Classification of Diseases for Oncology, ICSI: intracytoplasmic sperm injection, IL: Infant leukemias, IVF: in vitro fertilization, MLL: mixed-lineage leukemia, NHL: non-Hodgkin’s lymphoma, NSAID: Non-steroidal anti-inflammatory drug, PNET: Primitive Neuroectodermal Tumor, SEP: Socio-economic position, SNS: Sympathetic Nervous System.

## Web Figure 1. Example of Study Design Diagram




Web Figure 1: Example of study design diagram for diethylstilbestrol and vaginal adenocarcinoma (113). The box illustrating follow-up has been truncated to increase readability.

## Web Figure 2. Example of Directed Acyclic Graph




Web Figure 2: Example of directed acyclic graph for diethylstilbestrol and vaginal adenocarcinoma (114). Arrows between covariates are not displayed for reasons of readability. The white circles represent potential confounders, the yellow circle represent exposure, the blue circle with an "I" represents the outcome, the other blue circles represent intermediate variables. The green arrows indicate the causal pathways.

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